



Elliptic PDEs: Theory and Applications

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Vorwort

Dies sind die Notizen zu der Vorlesung „Elliptic PDEs: Theory and applications“. Sie werden vermutlich mit Fehlern gespickt sein und deshalb mit Vorsicht zu gebrauchen (und sind vorallem noch *nicht* fertiggestellt). Die Notizen basieren ungefähr auf der folgenden Literatur:

- [1] Serena Dipierro and Enrico Valdinoci. *Elliptic partial differential equations from an elementary viewpoint*. Jan. 31, 2025. DOI: 10.48550/arXiv.2101.07941. arXiv: 2101.07941[math]. URL: <http://arxiv.org/abs/2101.07941> (visited on 02/20/2025).
- [2] L.C. Evans. *Partial Differential Equations*. Graduate studies in mathematics. American Mathematical Society, 1998. ISBN: 978-0-8218-0772-9. URL: https://books.google.ch/books?id=5Pv4LVB_m8AC.
- [3] Xavier Fernández-Real and Xavier Ros-Oton. *Integro-Differential Elliptic Equations*. Vol. 350. Progress in Mathematics. Cham: Springer Nature Switzerland, 2024. ISBN: 978-3-031-54241-1 978-3-031-54242-8. DOI: 10.1007/978-3-031-54242-8. URL: <https://link.springer.com/10.1007/978-3-031-54242-8> (visited on 02/20/2025).
- [4] D. Gilbarg and N.S. Trudinger. *Elliptic Partial Differential Equations of Second Order*. Classics in Mathematics. Springer Berlin Heidelberg, 2001. ISBN: 978-3-540-41160-4. URL: <https://books.google.ch/books?id=eoIGTf4cmhwC>.

0 Introduction

Definition 0.1. Let $\Omega \subseteq \mathbb{R}^n$ be an open set. We call $u \in C^2(\Omega)$ **harmonic** if

$$\Delta u := \sum_{i=1}^n \partial_{i,i} u = 0$$

The space of harmonic functions form a vector space over $\mathbb{R}$.

1 Harmonic functions

1.1 Mean value property

Proposition 1.1. *Let $\Omega \subseteq \mathbb{R}^n$ a domain. For any $u \in C^2(\Omega)$ we have that*

$$\lim_{r \searrow 0} \frac{1}{r^2} \int_{B_r(x)} u(y) - u(x) \, dy = \frac{1}{2(n+2)} \Delta u(x).$$

Proof. Without loss of generality assume $x = 0$. By Taylor expansion we find

$$\begin{aligned} \frac{1}{r^2} \int_{B_r(0)} u(y) - u(0) \, dy &= \frac{1}{r^2} \int_{B_r(0)} \nabla u(0) \cdot y + y^\top \text{Hess}[u](0)y + o(|y|^2) \, dy \\ &= \frac{1}{r^2} \int_{B_r(0)} y^\top \text{Hess}[u](0)y \, dy + o(1) \\ &= \frac{1}{r^2} \sum_{i,j=1}^n \partial_{i,j}^2 u(0) \underbrace{\int_{B_r(0)} y_i y_j \, dy}_{=0 \text{ if } i \neq j} \\ &= \frac{1}{r^2} \sum_{i=1}^n \partial_i^2 u(0) \int_{B_r(0)} y_i^2 \, dy + o(1) \\ &= \frac{1}{2(n+2)} \Delta u(0) + o(1) \end{aligned}$$

for $r \searrow 0$. The value $\frac{1}{2(n+2)}$ comes from

$$\begin{aligned} \int_{B_r(0)} y_i^2 \, dy &= \int_{B_r(0)} \frac{1}{n} \sum_{j=1}^n y_j^2 \, dy = \frac{1}{|B_1(0)| r^n} \int_{B_r(0)} |y|^2 \, dy \\ &= \frac{1}{n |B_1(0)| r^n} \int_0^r \mathcal{H}^{n-1}(\partial B_1(0)) \rho^2 \rho^{n-1} \, d\rho \\ &= \frac{\mathcal{H}^{n-1}(\partial B_1(0))}{n |B_1(0)|} \cdot \frac{r^2}{n+2} = \frac{r^2}{n+2}. \end{aligned}$$

This concludes the proof. ♣

It turns out that a similar property holds for the average over the shell of the ball.

Proposition 1.2. *For any $u \in C^2(\Omega)$ it holds that*

$$\lim_{r \searrow 0} \frac{1}{r^2} \int_{\partial B_r(x)} u(y) - u(x) \, d\mathcal{H}_y^{n-1} = \frac{1}{2n} \Delta u(x).$$

Proof. The proof is in a similar fashion to the previous one. ♣

Theorem 1.3 (Mean value property of harmonic functions). *Let $u \in C^2(\Omega)$. The following are equivalent:*

- (i) $\Delta u = 0$ in Ω
- (ii) $u(x) = \int_{\partial B_r(x)} u(y) d\mathcal{H}_y^{n-1}$
- (iii) $u(x) = \int_{B_r(x)} u(y) dy$ for any $B_r(x) \Subset \Omega$.

Proof. First we consider the function

$$\phi(r) := \int_{\partial B_r(x)} u(y) d\mathcal{H}_y^{n-1} = \int_{\partial B_1} u(x + rz) d\mathcal{H}_z^{n-1}, \quad \text{for } r > 0.$$

The derivative of ϕ is given by

$$\begin{aligned} \phi'(r) &= \frac{d}{dr} \left(\int_{\partial B_1} u(x + rz) d\mathcal{H}_z^{n-1} \right) = \int_{\partial B_1} \nabla u(x + rz) \cdot z d\mathcal{H}_z^{n-1} \\ &= \int_{\partial B_r(x)} \nabla u(y) \cdot \underbrace{\frac{y-x}{r}}_{\text{outer normal}} d\mathcal{H}_y^{n-1} = \int_{\partial B_r(x)} \nabla u(y) \cdot \nu(y) d\mathcal{H}_y^{n-1} \\ &= \frac{1}{\mathcal{H}^{n-1}(\partial B_r(x)) r^{n-1}} \int_{B_r(x)} \underbrace{\operatorname{div}(\nabla u)}_{=\Delta u} dy. \end{aligned}$$

From this formula, we obtain the equivalence of (i) and (ii).

Assume that (ii) holds. Then

$$\begin{aligned} \int_{B_r(x)} u(y) dy &= \int_0^r \int_{\partial B_\rho(x)} u(y) d\mathcal{H}_y^{n-1} d\rho = \int_0^r \mathcal{H}^{n-1}(\partial B_\rho) u(x) d\rho \\ &= u(x) \mathcal{H}^{n-1}(\partial B_1) \int_0^r \rho^{n-1} d\rho = u(x) \underbrace{\mathcal{H}^{n-1}(\partial B_1) \frac{r^n}{n}}_{=|B_r|}. \end{aligned}$$

Now assume (iii). Then

$$\begin{aligned} \int_{\partial B_r(x)} u(y) dy &= \frac{d}{dr} \int_{B_r(x)} u(y) dy = \frac{d}{dr} (u(x) r^n |B_1|) \\ &= \underbrace{nr^{n-1} |B_1|}_{=\mathcal{H}^{n-1}(\partial B_r)} u(x). \end{aligned}$$

♣

Theorem 1.4 (Smoothness of harmonic functions). *Let $u \in C^2(\Omega)$ be harmonic. Then u is infinitely often in the interior of Ω .*

In order to prove this theorem we need to introduce mollifiers. Let $\eta \in C_c^\infty(B_1)$ be radial, $\eta \geq 0$ and such that $\int_{B_1} \eta \, dx = 1$. We then set $\eta_\varepsilon(x) := \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right)$. Then

$$\int_{\mathbb{R}^n} \eta_\varepsilon \, dx = \int_{B_\varepsilon} \eta \, dx = \int_{B_1} \eta \, dx = 1.$$

Proof of Theorem 1.4. Define $u^\varepsilon(x) = u \star \eta_\varepsilon(x) = \int_{\Omega} \eta_\varepsilon(x-y) \, dy$. We claim that this mollification is equal to u in $\Omega_\varepsilon = \{x \in \Omega \mid \text{dist}(x, \partial\Omega) > \varepsilon\}$. We have

$$\begin{aligned} u^\varepsilon(x) &= \int_{B_\varepsilon(x)} \eta_\varepsilon(x-y) u(y) \, dy \\ &= \int_{B_\varepsilon(x)} \frac{1}{\varepsilon^n} \eta\left(\frac{|x-y|}{\varepsilon}\right) u(y) \, dy = \frac{1}{\varepsilon^n} \int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right) \int_{\partial B_r(x)} u(y) \, d\mathcal{H}_y^{n-1} \, dr \\ &\stackrel{\text{Mean value prop}}{=} \frac{1}{\varepsilon^n} \int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right) \int_{\partial B_r(x)} u(x) \, d\mathcal{H}_y^{n-1} \, dr \\ &= u(x) \int_{B_\varepsilon(x)} \eta_\varepsilon(x-y) \, dy = u(x), \end{aligned}$$

so $u^\varepsilon = u \in C^\infty(\Omega_\varepsilon)$ for any $\varepsilon > 0$. ♣

1.2 Local estimates

Theorem 1.5. *It holds that*

$$|\partial^\alpha u(x)| \lesssim_{n, |\alpha|} \frac{1}{r^{n+|\alpha|}} \|u\|_{L^1(B_r)}.$$

where α is a multiindex and u harmonic in B_r .

Proof. We write $k = |\alpha|$. If $k = 0$. Then the claim follows from the mean value property. If $k = 1$, we want to control $|\partial_j u|$ which is also a harmonic function. In fact we have

$$\partial_j u(x) = \frac{1}{|B_{r/2}|} \int_{B_{r/2}} \partial_j u(y) \, dy = \frac{1}{|B_1|} \left(\frac{1}{n}\right)^n \int_{\partial B_{r/2}} u \nu_j \, d\mathcal{H}^{n-1}.$$

Taking absolute values we obtain the estimate

$$|\partial_j u(x)| \leq \frac{1}{|B_1|} \left(\frac{1}{n}\right)^n \|u\|_{L^\infty(B_{r/2})} \mathcal{H}^{n-1}(\partial B_{r/2}) \leq \frac{2n}{r} \|u\|_{L^\infty(\partial B_{r/2})}$$

$$\leq \frac{\overbrace{2nC_0}^{C_1}}{r^{n+1}} \|u\|_{L^1(B_r)}.$$

Inductively, since $\Delta(\partial^\alpha u) = 0$,

$$\begin{aligned} |\partial_j \partial^\alpha u(x)| &\leq \frac{n(k+1)}{r} \|\partial^\alpha u\|_{L^\infty(\partial B_{\frac{r}{k+1}}(x))}, \quad \text{as in } k=1 \\ &\leq \frac{n(k+1)}{r} \frac{C_k}{\frac{k}{k+1} r^{n+k}} \|u\|_{L^1(B_r(x))} \\ &\leq \underbrace{\frac{n(k+1)^2}{k} C_k}_{=: C_{k+1}} \frac{1}{r^{n+k+1}} \|u\|_{L^1(B_r(x))}. \end{aligned}$$

This concludes the induction step and we are done. ♣

Corollary 1.6. *If u is harmonic in Ω then u is analytic.*

Proof. The proof is left as an exercise to the reader or see [2]. ♣

Note in the one dimensional case, $f(x) = \sum_{k=0}^{+\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$. If $|f^{(k)}(x_0)| \leq C_0^{k+1} k!$ then the series converges for $|x - x_0| \leq \frac{1}{x_0^2}$.

Example 1.7. • By complex analysis, the real and imaginary parts of a holomorphic function are harmonic on $\mathbb{R}^2$.

- Constants in $\mathbb{R}^n$, linear functions in $\mathbb{R}^n$ are harmonic.
- In $\mathbb{R}^2$, $u(x, y) = x^2 - y^2$ and $u(x, y) = 2xy$ are harmonic. These come from taking the real and imaginary part of $f : \mathbb{C} \rightarrow \mathbb{C}$, $z \mapsto z^2$ (check with the first point). Same works for $u(x, y) = e^x \cos(y)$ and $u(x, y) = e^x \sin y$. Also one can take higher powers z^k , $z \in \mathbb{C}$.

♦

1.3 Fundamental solution of the Laplacian

We seek a radial function that is harmonic. Assuming that $u = u(|x|)$ we have, writing $r = |x|$,

$$\begin{aligned} \Delta u &= \sum_{j=1}^n \partial_j^2 (u(r)) = \sum_j \left[u''(r) \left(\frac{x_j}{r} \right)^2 + u'(r) \left(\frac{1}{r} - \frac{x_j}{r^3} \right) \right] \\ &= u''(r) + \frac{n-1}{r} u'(r) \stackrel{(!)}{=} 0 \end{aligned}$$

So we only need to solve an ODE. Rewriting this yields

$$r^{1-n}(r^{n-1}u')' = 0 \implies r^{n-1}u' = C \implies u' = C_1 r^{1-n}$$

which leads to

$$u = \begin{cases} C_1 \log r + C_2 & n = 2 \\ C_3 r^{2-n} + C_2 & n \geq 3. \end{cases}$$

1.4 Maximum principle

Theorem 1.8. Suppose $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ and $\Delta u = 0$ in Ω , where Ω is bounded. Then:

- (i) $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u$;
- (ii) If $u(x) = \max_{\overline{\Omega}} u$ for x in the interior point of $x \in \Omega$ and Ω is connected, then $u = \text{const}$ on Ω .

Remark 1.9. The same holds for the minimum by considering $-u$. ♦

Proof of Theorem 1.8. Suppose that the maximum of u , denoted by M , is attained at $x \in \Omega$ then for $B_r(x) \subseteq \Omega$,

$$M = u(x) = \int_{B_r(x)} u(y) \, dy \leq M$$

so $u \equiv M$ on $B_r(x)$. Hence $\{x \in \Omega \mid u(x) = M\}$ is both open (by the above) and relatively closed (by definition), i.e., it equals Ω if Ω is connected. Now observe that the second property implies the first so we are done. ♣

1.5 Uniqueness and Liouville theorem (classification)

Corollary 1.10. Let Ω be a **bounded** domain. If u satisfies

$$\begin{cases} -\Delta u = 0 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

then $u \equiv 0$.

Proof. We have $\max_{\overline{\Omega}} u = \max_{\partial\Omega} u = 0$ and the analogous result holds for $-\min_{\overline{\Omega}} u$. So $u \equiv 0$. ♣

Theorem 1.11. If $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is harmonic and bounded, then $u \equiv \text{const}$.

Proof. For any $x \in \mathbb{R}^n$,

$$|\nabla u(x)| \leq \frac{C_1}{r^{n+1}} \|u\|_{L^1(B_r(x))} \leq \frac{C_1 |B_1|}{r} \|u\|_{L^\infty(\mathbb{R}^n)} \leq \frac{C}{r} \xrightarrow{r \rightarrow +\infty} 0. \quad \clubsuit$$

The maximum principle rules out isolated zeros. Another way to see this is Harnack's inequality.

Theorem 1.12. *If $V \Subset \Omega$ (is compactly contained in Ω) is connected, then for any non-negative (on Ω) harmonic function we have*

$$\sup_V u \lesssim_V \inf_V u$$

or equivalently

$$C_V^{-1} u(y) \leq u(x) \leq C_V u(y) \quad \text{for any } x, y \in \Omega$$

with a suitable constant $C_V > 0$, dependent on V .

Proof. Let $r = \frac{1}{4} \text{dist}(V, \partial\Omega)$. Take $x, y \in V$ with $|x - y| \leq r$, $B_r(y) \subseteq B_{2r}(x)$ so

$$\begin{aligned} u(x) &= \int_{B_{2r}(x)} u(z) \, dz = \frac{1}{|B_r|(2r)^n} \int_{B_{2r}(x)} u(z) \, dz \\ &\geq \frac{1}{|B_1|(2r)^n} \int_{B_r(y)} u(z) \, dz, \quad \text{using } u \geq 0 \\ &= \frac{1}{2^n} u(y). \end{aligned}$$

Since V is connected and $\bar{V}$ is compact we can cover $\bar{V}$ by a chain of finitely many balls $\{B_i\}_{i=1}^N$ with $N = N(V)$ and $B_i \cap B_{i+1} \neq \emptyset$. Then $u(x) \geq \frac{1}{2^{nN}} u(y)$. $\clubsuit$

2 Poisson's equation

We now turn to Poisson's equation

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega \end{cases} \quad (2.1)$$

where $\Omega \subseteq \mathbb{R}^n$ is either bounded or a halfspace. This is again a linear PDE so to check uniqueness it is enough to show that $u \equiv 0$ is the only solution if $f \equiv 0$, $g \equiv 0$. However we already discussed this case in the previous sections.

Given $f \in C(\Omega)$, $g \in C(\partial\Omega)$ what is u in terms of f and g ?

The goal is to prove the formula

$$u(x) = \int_{\Omega} G(x, y) f(y) \, dy + \int_{\partial\Omega} P(x, y) g(y) \, d\mathcal{H}_y^{n-1}.$$

Recall Green's second identity: for $u, G \in C^2$

$$\int_{\Omega} (u \Delta G - G \Delta u) \, dx = \int_{\partial\Omega} u \partial_{\nu} G - G \partial_{\nu} u \, d\mathcal{H}^{n-1}.$$

This is a consequence of Greens first identity:

$$\int_{\Omega} (u \Delta G + \nabla u \cdot \nabla G) \, dy = \int_{\partial\Omega} u \partial_{\nu} G \, d\mathcal{H}^{n-1}$$

which is just a version of the divergence theorem. As another motivation consider

Lemma 2.1. *Let $u \in C(\Omega)$ and $x \in \Omega \setminus \{0\} \subseteq \mathbb{R}^n$. Then*

$$\int_{\Omega} u(y) \eta_{\varepsilon}(y - x) \, dy \xrightarrow{\varepsilon \searrow 0} u(x)$$

where η_{ε} is a standard Friederich's mollifier of the form $\eta_{\varepsilon} := \frac{1}{\varepsilon^n} \eta\left(\frac{\cdot}{\varepsilon}\right)$.

Proof. Using dominated convergence theorem, one gets

$$\begin{aligned} \int_{\Omega} u(y) \eta_{\varepsilon}(x - y) \, dy &= \int_{B_{\varepsilon}(x)} u(y) \frac{1}{\varepsilon^n} \eta\left(\frac{|x - y|}{\varepsilon}\right) \, dy \\ &= \int_{B_1} u(x + \varepsilon z) \eta(|z|) \, dz \xrightarrow{\varepsilon \searrow 0} u(x) \int_{B_1} \eta(|z|) \, dz = u(x). \quad \clubsuit \end{aligned}$$

Formally we write $\eta_{\varepsilon} \rightarrow \delta_x$ i.e.,

$$\int_{\Omega} u(y) \eta_{\varepsilon}(x - y) \, dy \xrightarrow{\varepsilon \searrow 0} \int_{\Omega} u(y) \, d\delta_x(y).$$

Proposition 2.2. Let $\varepsilon > 0$. For $n \geq 3$ define $\Phi^\varepsilon \in C^2(\mathbb{R}^n)$ by

$$\Phi^\varepsilon(x) = \begin{cases} (|x|^{n-2}(n-2)\mathcal{H}^{n-1}(\partial B_1))^{-1} & |x| \geq \varepsilon \\ a_\varepsilon - b_\varepsilon|x|^2 + c_\varepsilon|x|^4 & |x| < \varepsilon \end{cases}$$

where $a_\varepsilon, b_\varepsilon, c_\varepsilon \in \mathbb{R}$ are accordingly chosen. Then

1. $-\Delta\Phi^\varepsilon = 0$ on $\mathbb{R}^n \setminus B_\varepsilon$ and
2. $\int_{B_1} -\Delta\Phi^\varepsilon(y) dy = 1$.

Proof. The result follows by a calculation. In the second formula, one has to integrate by parts and then obtains the result. ♣

Therefore, one can think of $-\Delta\Phi^\varepsilon$ as converging to δ_0 as $\varepsilon \searrow 0$, by translation, $-\Delta\Phi^\varepsilon(x - \cdot)$ to δ_x . The only problem is that it is not zero on the boundary.

Definition 2.3. We define

$$\Phi_x(y) := \begin{cases} ((n-2)\mathcal{H}^{n-1}(\partial B_1)|x-y|^{n-2}), & n \geq 3 \\ \frac{1}{2\pi} \log \frac{1}{|x-y|} & n = 2 \end{cases}$$

and call it **fundamental solution of the Laplacian with center x** .

We can obtain Green's function from the fundamental solution. Fix $x \in \Omega$. We know that $-\Delta\Phi_x = \delta_x$ in Ω . We want

$$\begin{cases} -\Delta G_x = \delta_x & \text{in } \Omega \\ G = 0 & \text{on } \partial\Omega \end{cases}.$$

So we subtract and obtain

$$\begin{cases} -\Delta(\Phi_x - G_x) = 0 & \text{in } \Omega \\ \Phi_x - G_x = \Phi_x & \text{on } \partial\Omega \end{cases}.$$

We define the corrector ϕ_x as the solution of

$$\begin{cases} -\Delta\phi_x = 0 & \text{on } \Omega \\ \phi_x = \Phi_x & \text{on } \partial\Omega \end{cases}. \quad (2.2)$$

Then $G(x, y) = G_\Omega(x, y) = G_x(y) = \Phi_x(y) - \phi_x(y)$.

Theorem 2.4. Let $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ solves (2.1) then

$$u(x) = \int_{\Omega} G(x, y)f(y) dy + \int_{\partial\Omega} -\partial_{n(y)}G(x, y)g(y) d\mathcal{H}_y^{n-1},$$

where G is choosen as above and $\partial_{n(y)}G(x, y) = \nabla_y G(y) \cdot \nu(y)$.

Proof. We integrate by parts on $\Omega \setminus B_\varepsilon(x)$ for some $\varepsilon > 0$

$$\begin{aligned} \int_{\Omega \setminus B_\varepsilon(x)} (u \Delta G - G \Delta u) dy &= \int_{\partial(\Omega \setminus B_\varepsilon(x))} (u \partial_\nu G - G \partial_\nu u) d\mathcal{H}^{n-1} \\ &= \int_{\partial\Omega} (u \partial_\nu G - G \partial_\nu u) d\mathcal{H}^{n-1} - \int_{\partial B_\varepsilon(x)} (u \partial_\nu G - G \partial_\nu u) d\mathcal{H}^{n-1} \end{aligned}$$

Note the negative-sign. This is because the outward vector of $\partial(\Omega \setminus B_\varepsilon(x))$ points inside the ball $B_\varepsilon(x)$. Moreover, for $\varepsilon \searrow 0$,

$$\begin{aligned} - \int_{\partial B_\varepsilon(x)} u \partial_\nu G d\mathcal{H}^{n-1} &= - \int_{B_\varepsilon(x)} u(y) \partial_{\nu(y)} \Phi(y-x) d\mathcal{H}_y^{n-1} + \int_{\partial B_\varepsilon(x)} \underbrace{u(y) \partial_{\nu(y)} \phi^x(y)}_{\text{bounded}} d\mathcal{H}_y^{n-1} \\ &= - \int_{\partial B_\varepsilon(x)} u(y) \frac{y-x}{\varepsilon} \cdot \nabla_y \left(\frac{|y-x|^{2-n}}{(n-2)\mathcal{H}^{n-1}(\partial B_1)} \right) d\mathcal{H}_y^{n-1} + o(\varepsilon^{n-1}) \\ &= \int_{\partial B_\varepsilon(x)} u(y) \frac{|y-x|^{2-n}}{\varepsilon \mathcal{H}^{n-1}(\partial B_1)} d\mathcal{H}_y^{n-1} + o(\varepsilon^{n-1}) \\ &= \int_{\partial B_\varepsilon(x)} u(y) d\mathcal{H}_y^{n-1} + o(\varepsilon^{n-1}) \xrightarrow{\varepsilon \searrow 0} u(x) \end{aligned}$$

and

$$\begin{aligned} - \int_{B_\varepsilon(x)} -G \partial_\nu u d\mathcal{H}^{n-1} &= \int_{\partial B_\varepsilon(y)} \frac{|y-x|^{2-n}}{(n-2)\mathcal{H}^{n-1}(\partial B_1)} \partial_\nu u(y) d\mathcal{H}_y^{n-1} \quad (2.3) \\ &= o(e^{2-n}\mathcal{H}^{n-1}(\partial B_1)) = o(\varepsilon) \rightarrow 0 \end{aligned}$$

which shows that

$$\lim_{\varepsilon \searrow 0} - \int_{\partial B_\varepsilon(x)} (u \partial_\nu G - G \partial_\nu u) d\mathcal{H}^{n-1} = u(x).$$

This concludes the proof. ♣

Using similar arguments we can show that the Green's function is symmetric.

Theorem 2.5. For any $x, y \in \Omega$ with $x \neq y$ it holds $G(x, y) = G(y, x)$.

Proof. Let $\varepsilon > 0$ such that $B_\varepsilon(x)$ and $B_\varepsilon(y)$ do not intersect and integrate by parts in $\Omega \setminus (B_\varepsilon(x) \cup B_\varepsilon(y))$. We thus obtain,

$$0 = \int_{\Omega \setminus (B_\varepsilon(x) \cup B_\varepsilon(y))} G_x \Delta G_y - G_y \Delta G_x dz = \int_{\partial(\Omega \setminus (B_\varepsilon(x) \cup B_\varepsilon(y)))} \underbrace{G_x \partial_\nu G_y}_{=0 \text{ on } \partial\Omega} - \underbrace{G_y \partial_\nu G_x}_{=0 \text{ on } \partial\Omega} d\mathcal{H}^{n-1}$$

so we have

$$\int_{\partial B_\varepsilon(x)} \underbrace{G_x \partial_\nu G_y}_{o(\varepsilon^{2-n}) \text{ bounded}} - G_y \partial_\nu G_x \, d\mathcal{H}^{n-1} = \int_{\partial B_\varepsilon(y)} G_y \partial_\nu G_x - G_x \partial_\nu G_y \, d\mathcal{H}^{n-1}$$

letting $\varepsilon \searrow 0$ we obtain $G(x, y) = G(y, x)$. ♣

If we want to find Green's function we need to solve for the corrector given by (2.2). For domains with simple geometry it is possible to reflect away the singularity. The halfspace $\mathbb{H}^n$ is given by

$$\mathbb{H}^n = \{(x', x_n) \in \mathbb{R}^{n-1} \times \mathbb{R} \mid x_n > 0\}.$$

The reflection of a point $(x', x_n) \in \mathbb{R}^n$ is given by $\tilde{x} = (x, -x_n)$. Further, $\phi_x(y) = \Phi(y - \tilde{x})$ satisfies

$$\begin{cases} -\Delta \phi_x(y) = 0 & \text{in } \mathbb{R}^n \setminus \{\tilde{x}\} \supseteq \mathbb{H}^n \\ \phi_x(y) = \Phi(y - \tilde{x}) = \Phi(y - x) & \text{on } \partial\mathbb{H}^n \end{cases}.$$

From this we can write down the Green's function:

$$G(x, y) = \Phi(y - x) - \Phi(y - \tilde{x}) = \frac{1}{(n-2)\mathcal{H}^{n-1}(\partial B_1)} \left(\frac{1}{|x - y|^{n-1}} - \frac{1}{|y - \tilde{x}|^{n-2}} \right).$$

For $u \in C^2(\bar{\Omega})$ recall the representation

$$u(x) = \int_{\Omega} G(x, y) f(y) \, dy + \int_{\partial\Omega} -\partial_{\nu(y)} G(x, y) g(y) \, dy$$

for (2.1) with $\Omega = \mathbb{H}^n$. In this case we call $-\partial_{\nu(y)} G(x, y)$ **Poisson kernel**. Here we have $\partial_\nu = -\partial_{y_n}$ so

$$-\partial_\nu G(x, y) = \frac{-1}{\mathcal{H}^{n-1}(\partial B_1)} \left(\frac{y_n - x_n}{|y - x|^n} - \frac{y_n - \tilde{x}_n}{|y - \tilde{x}|^n} \right).$$

For $y \in \partial\mathbb{H}^n$,

$$-\partial_{\nu(y)} G(x, y) = \frac{2x_n}{\mathcal{H}^{n-1}(\partial B_1)|y - x|^n}.$$

Theorem 2.6 (Poisson's Formula). *Let $g \in C^0(\partial\mathbb{H}^n) \cap L^\infty(\partial\mathbb{H}^n)$. Then*

$$u(x) = \int_{\partial\mathbb{H}^n} P(x, y) g(y) \, dy$$

satisfies:

$$(1) \ u \in C^\infty \cap L^\infty(\mathbb{H}^n),$$

(2) $\Delta u = 0$ on $\mathbb{H}^2$ and

(3) $\lim_{\mathbb{H}^n \ni x \rightarrow x^0} u(x) = g(x^0)$.

Proof. (1) See [2]. To show that $u \in L^\infty(\mathbb{H}^n)$, we need

$$\int_{\partial\mathbb{H}^n} P(x, y) dy = \frac{2x_n}{\mathcal{H}^{n-1}(\partial B_1)} \int_{\partial\mathbb{H}^n} \frac{dy}{|x - y|^n} = 1.$$

Indeed,

$$\begin{aligned} \int_{\partial\mathbb{H}^n} \frac{2x_n}{|x - y|^n} dy &= \int_{\mathbb{R}^{n-1}} \frac{2x_n}{(|x' - y'|^2 + x_n^2)^{n/2}} dy' = \int_{\mathbb{R}^{n-1}} \frac{2x_n dy'}{(|y'|^2 + x_n^2)^{n/2}} \\ &= \int_{\mathbb{R}^{n-1}} \frac{2 dz'}{(|z'|^2 + 1)^{n/2}} = 2\mathcal{H}^{n-2}(\mathbb{S}^{n-2}) \int_0^{+\infty} \frac{r^{n-2} dr}{(r^2 + 1)^{n/2}}. \end{aligned}$$

Substituting $r = \tan \theta$, i.e., $r^2 + 1 = \sec^2 \theta$ and $dr = \sec^2 \theta d\theta$ we obtain

$$\begin{aligned} \int_0^{+\infty} \frac{r^{n-2} dr}{(r^2 + 1)^{n/2}} &= \int_0^{\pi/2} \frac{(\tan \theta)^{n-2} (\sec \theta)^2}{(\sec \theta)^n} d\theta = \int_0^{\pi/2} \sin^{n-1} \theta d\theta \\ &= \frac{1}{2} \int_0^\pi \sin^{n-2}(\theta) d\theta = \frac{1}{2\mathcal{H}^{n-2}(\mathbb{S}^{n-2})} \int_{\mathbb{S}^{n-1}} d\mathcal{H}^{n-1} \\ &= \frac{1}{2\mathcal{H}^{n-2}(\mathbb{S}^{n-2})} \mathcal{H}^{n-1}(\partial B_1) \end{aligned}$$

so the integral is equals to one as we wanted. Now we have

$$|u(x)| \leq \|g\|_{L^\infty(\partial\mathbb{H}^n, \mathcal{H}^{n-1})} \int_{\partial\mathbb{H}^n} \frac{2x_n}{\mathcal{H}^{n-1}(\partial B_1) |x - y|^n} \leq \|g\|_{L^\infty(\partial\mathbb{H}^n, \mathcal{H}^{n-1})},$$

also, $u \in C^\infty(\mathbb{H}^n)$ since Poisson's Kernel is C^∞ or because u is harmonic in $\mathbb{H}^n$.

(2) Since $P(x, y)$ is regular ($x \in \mathbb{H}^n$ and $y \in \partial\mathbb{H}^n$ are distinct), it is harmonic. So

$$-\Delta u(x) = \int_{\partial\mathbb{H}^n} -\Delta_x P(x, y) g(y) dy = 0.$$

(3) Fix $x^0 \in \partial\mathbb{H}^n$. For $\varepsilon > 0$, take $\delta > 0$ such that

$$|g(y) - g(x^0)| < \varepsilon, \quad \text{if } |y - x^0| < \delta, \quad y \in \partial\mathbb{H}^n.$$

Then if $x \in \mathbb{H}^n$ with $|x^0 - x| < \delta/2$,

$$\begin{aligned} |u(x) - g(x^0)| &= \left| \int_{\partial\mathbb{H}^n} P(x, y) (g(y) - g(x^0)) dy \right| \\ &\leq \int_{\partial\mathbb{H}^n \cap B_\delta(x^0)} P(x, y) \underbrace{|g(y) - g(x^0)|}_{\leq \varepsilon} dy + C(n) \underbrace{x_n}_{\rightarrow 0} \int_{\partial\mathbb{H}^n \setminus B_\delta(x^0)} \frac{|g(y) - g(x^0)|}{|x^0 - y|^n} dy \end{aligned}$$

$$\begin{aligned}
&\leq \varepsilon \int_{\partial \mathbb{H}^n \cap B_\delta(x_0)} P(x, y) \, dy \\
&\quad + C(n) x_n \int_{\partial \mathbb{H}^n \setminus B_\delta(x^0)} \left(\frac{2}{|x^0 - y|} \right)^n 2 \|g\|_{L^\infty(\partial \mathbb{H}^n, \mathcal{H}^{n-1})} \, dy \\
&\leq 2\varepsilon
\end{aligned}$$

if $|x - x^0|$ is small enough.

♣

3 Energy methods (involving L^1 or H^1 -norms)

Using energy methods we can give an alternative proof of the uniqueness of the solution to Poisson's equation

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega. \end{cases} \quad (3.1)$$

Proof. Let $u, \tilde{u}$ be solutions of (3.1). Then $w := u - \tilde{u}$ solves

$$\begin{cases} -\Delta u = 0 & \text{in } \Omega \\ w = 0 & \text{on } \partial\Omega. \end{cases}$$

Testing against w , that is, multiplying both sides by w and integrating by parts on Ω , we obtain

$$0 = - \int_{\Omega} w \Delta w \, dx = \int_{\Omega} |\nabla w|^2 \, dx.$$

Note that the boundary integral vanishes because of the zero boundary conditions on w . Thus $\nabla w \equiv 0$ and thus w is constant. By the boundary conditions $w \equiv 0$ which concludes the proof. ♣

We now introduce the Dirichlet Principle which says, that u solves the PDE if and only if u minimizes the energy. The energy is given by

$$E[v] = \int_{\Omega} \frac{1}{2} |\nabla v|^2 - v f \, dx$$

where f is given by (3.1). We define the set of admissible functions (over which we will minimize) as

$$\mathcal{A} := \left\{ v \in C^2(\bar{\Omega}) \mid v = g \text{ on } \partial\Omega \right\}.$$

Theorem 3.1. $u \in C^2(\bar{\Omega})$ solves (3.1) if and only if

$$u = \operatorname{argmin}_{v \in \mathcal{A}} E[v]. \quad (3.2)$$

Proof. Necessity of (3.2): Let $w \in \mathcal{A}$. We want $E[u] \leq E[w]$. Testing against $u - w$ and integrating by parts,

$$0 = \int_{\Omega} (-\Delta u - f)(u - w) \, dx = \int_{\Omega} |\nabla u|^2 - f u \, dx - \int_{\Omega} \nabla u \cdot \nabla w - f w \, dx$$

thus,

$$E[u] = \int_{\Omega} \frac{1}{2} |\nabla u|^2 - f u \, dx = \int_{\Omega} \underbrace{\frac{\nabla u \cdot \nabla w}{\frac{1}{2} |\nabla u|^2 + \frac{1}{2} |\nabla w|^2}}_{\leq \frac{1}{2} |\nabla u|^2 + \frac{1}{2} |\nabla w|^2} - f w - \frac{1}{2} |\nabla u|^2 \, dx$$

$$\leq \int_{\Omega} \frac{1}{2} |\nabla w|^2 - fw \, dx = E[v].$$

Sufficiency of (3.2): Let $u \in \mathcal{A}$ and $E[u] \leq E[w]$ for any $w \in \mathcal{A}$. For any $v \in C_c^\infty(\Omega)$ and $t > 0$, $u + tv = g$ on $\partial\Omega$. So $u + tv \in \mathcal{A}$ and we have

$$\begin{aligned} E[u + tv] &= \int_{\Omega} \frac{1}{2} |\nabla u + t\nabla v|^2 - (u + tv)f \, dx \\ &= \int_{\Omega} \frac{1}{2} |\nabla u|^2 + t\nabla u \cdot \nabla v + \frac{t^2}{2} |\nabla v|^2 - uf - tvf \, dx. \end{aligned}$$

Since $E[u + tv]$ attains its minimum at $t = 0$,

$$0 = \left. \frac{d}{dt} E[u + tv] \right|_{t=0} = \int_{\Omega} \nabla u \cdot \nabla v - vf \, dx = \int_{\Omega} (-\Delta u - f)v \, dx.$$

By the arbitrariness of $v \in C_c^\infty(\Omega)$ we conclude. ♣

Hence, to show that the PDE has a solution, we solve a minimization problem. The “correct” space is the **Sobolev space**, which is just enough to make sense of the integral of (weak) derivatives in L^p . We define, for $p \in [1, +\infty]$,

$$W^{1,p}(\Omega) := \{u \in L^p(\Omega) \mid \partial_j u \in L^p \forall j = 1, \dots, n\},$$

where $\partial_j u$ denotes the weak derivative of u . We endow this space with the norm

$$\|u\|_{W^{1,p}(\Omega)} = \sqrt[p]{\int_{\Omega} |u|^p + |\nabla u|^p \, dx}, \quad \text{for } p \in [1, +\infty).$$

In the following, we give a list of useful properties of those Sobolev spaces. We will not prove these.

- (S1) $W^{1,p}(\Omega)$ is complete.
- (S2) The inclusion $W^{1,p}(\Omega) \subseteq L^p(\Omega)$ is compact.
- (S3) $H^1(\Omega) := W^{1,2}(\Omega)$ is a Hilbert space with the inner product

$$\langle u, v \rangle_{H^1(\Omega)} = \int_{\Omega} uv + \nabla u \cdot \nabla v \, dx.$$

- (S4) Any bounded sequence $\{u_k\}_{k \in \mathbb{N}} \subseteq H^1(\Omega)$ contains a weakly convergent subsequence $\{u_{k_j}\}_{j \in \mathbb{N}}$. Moreover this since $H^1(\Omega) \subseteq L^2(\Omega)$, we have

$$\|u\|_{L^2(\Omega)} = \lim_{j \rightarrow +\infty} \|u_{k_j}\|_{L^2(\Omega)},$$

i.e., every bounded sequence in $H^1(\Omega)$ has a convergent subsequence.

(S5) For Ω bounded Lipschitz domain and $p \in [1, +\infty)$, the trace (boundary) operator

$$T : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega, \mathcal{H}^{n-1}), \quad u \mapsto u \Big|_{\partial\Omega} \quad \text{if } u \in C^1(\Omega) \cap W^{1,p}(\Omega)$$

is continuous (and compact for $1 < p < +\infty$):

$$\|Tu\|_{L^p(\partial\Omega)} \lesssim \|u\|_{W^{1,p}(\Omega)}.$$

We will also use the notation $u|_{\partial\Omega} := Tu$.

(S6) $W_0^{1,p}(\Omega)$ is the subspace,

$$\left\{ u \in W^{1,p}(\Omega) \mid u \Big|_{\partial\Omega} \equiv 0 \right\}.$$

(S7) Let $p \in [1, +\infty)$. Then $C^\infty(\overline{\Omega})$ -functions are dense in $W^{1,p}(\Omega)$ and $C_c^\infty(\Omega)$ -functions are dense in $W_0^{1,p}(\Omega)$.

(S8) Let $p \in [1, +\infty]$ and Ω a bounded Lipschitz domain. Then there exists a well-defined extension operator

$$E : W^{1,p}(\Omega) \rightarrow W(1,p)(\mathbb{R}^n)$$

which is continuous and

$$\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \lesssim \|u\|_{W^{1,p}(\Omega)}.$$

(S9) If $u \in W^{1,p}(\Omega)$ for $p \in [1, +\infty]$, then for any compact set $K \subseteq \Omega$ and $|h| \leq \delta$,

$$\left\| \frac{u(\cdot + h) - u}{h} \right\|_{L^p(K)} \lesssim \|\nabla u\|_{L^p(\Omega)}.$$

Conversly if $u \in L^p(\Omega)$ for $p \in (1, +\infty]$ and

$$\left\| \frac{u(\cdot + h) - u}{h} \right\|_{L^p(K)} \leq C \quad \text{for all } |h| \leq \delta,$$

then $u \in W^{1,p}(K)$ and $\|u\|_{W^{1,p}(\Omega)} \leq C$.

(S10) For any $u \in W^{1,p}(\Omega)$, then $u^+, u^- \in W^{1,p}(\Omega)$ and $\nabla u = \nabla u^+ - \nabla u^-$ a.e. in Ω .

Theorem 3.2 (Sobolev inequality). *Let $1 \leq p < n$. Then*

$$\|u\|_{L^{p^*}(\mathbb{R}^n)} \lesssim_{p,n} \|\nabla u\|_{L^p(\mathbb{R}^n)}$$

where $p^* = \frac{np}{n-p}$ i.e., $\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}$.

Remark 3.3. The parameters can be seen by scaling. If the inequality holds for any $u \in C_c^1(\mathbb{R}^n)$ for some q instead of p^* , then it also holds for its rescaling

$$u_\lambda(x) := u(\lambda x).$$

Then by change of variables,

$$\|u\|_{L^q(\mathbb{R}^n)} = \lambda^{n/q} \|u_\lambda\|_{L^q} \leq C(n, p) \lambda^{n/q} \underbrace{\|\nabla u_\lambda\|_{L^p}}_{=\lambda^{1-n/p} \|\nabla u\|_{L^p}} = C(n, p) \lambda^{1+\frac{n}{p}+\frac{n}{q}} \|\nabla u\|_{L^p},$$

which indicates that $1 - \frac{n}{p} + \frac{n}{q} = 0$ should hold i.e., that our choice of p^* is correct. $\blacklozenge$

Proof. Step 1: The case $p = 1$. Since $u \in C_c^1(\mathbb{R}^n)$, we can write for each $i = 1, \dots, n$ and $x \in \mathbb{R}^n$,

$$u(x) = \int_{-\infty}^{x_i} \partial_{x_i} u(x_1, \dots, x_{i-1}, y_i, x_{i+1}, \dots, x_n) dy_i,$$

and thus

$$|u(x)| \leq \int_{\mathbb{R}} |\nabla u(x_1, \dots, y_i, \dots, x_n)| dy_i.$$

It follows that

$$|u(x)|^{\frac{n}{n-1}} \leq \prod_{i=1}^n \left(\int_{\mathbb{R}} |\nabla u(x_1, \dots, y_i, \dots, x_n)| dy_i \right)^{\frac{1}{n-1}}.$$

Integrate with respect to x_1 :

$$\begin{aligned} \int_{-\infty}^{\infty} |u(x)|^{\frac{n}{n-1}} dx_1 &\leq \int_{\mathbb{R}} \prod_{i=1}^n \left(\int_{\mathbb{R}} |\nabla u| dy_i \right)^{\frac{1}{n-1}} dx_1 \\ &= \left(\int_{\mathbb{R}} |\nabla u| dy_1 \right)^{\frac{1}{n-1}} \int_{\mathbb{R}} \prod_{i=2}^n \left(\int_{\mathbb{R}} |\nabla u| dy_i \right)^{\frac{1}{n-1}} \\ &\leq \left(\int_{-\infty}^{\infty} |\nabla u| dy_1 \right)^{\frac{1}{n-1}} \prod_{i=2}^n \left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |\nabla u| dx_1 dy_i \right)^{\frac{1}{n-1}}, \end{aligned}$$

the last inequality resulting from the general Hölder inequality.

Now integrate the above with respect to x_2 :

$$\iint_{\mathbb{R} \times \mathbb{R}} |u|^{\frac{n}{n-1}} dx_1 dx_2 \leq \left(\iint_{\mathbb{R} \times \mathbb{R}} |\nabla u| dx_1 dx_2 \right)^{\frac{1}{n-1}} \int_{\mathbb{R}} \prod_{\substack{i=1 \\ i \neq 2}}^n I_i^{\frac{1}{n-1}} dx_2,$$

where

$$I_1 = \int_{\mathbb{R}} |\nabla u| \, dy_1, \quad I_i = \iint_{\mathbb{R} \times \mathbb{R}} |\nabla u| \, dx_1 \, dy_i \quad \text{for } i = 3, \dots, n.$$

By applying the extended Hölder inequality once more, we find

$$\begin{aligned} & \int_{\mathbb{R}} \int_{\mathbb{R}} |u|^{\frac{n}{n-1}} \, dx_1 \, dx_2 \\ & \leq \left(\int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| \, dx_1 \, dy_2 \right)^{\frac{1}{n-1}} \left(\int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| \, dy_1 \, dx_2 \right) \prod_{i=3}^n \left(\int_{\mathbb{R}} \int_{\mathbb{R}} \int_{\mathbb{R}} |\nabla u| \, dx_1 \, dx_2 \, dy_i \right)^{\frac{1}{n-1}} \end{aligned}$$

We continue by integrating with respect to $x_3, \dots, x_n$, eventually to find

$$\begin{aligned} \int_{\mathbb{R}^n} |u|^{\frac{n}{n-1}} \, dx & \leq \prod_{i=1}^n \left(\int_{\mathbb{R}} \dots \int_{\mathbb{R}} |\nabla u| \, dx_1 \dots dy_i \dots dx_n \right)^{\frac{1}{n-1}} \\ & = \left(\int_{\mathbb{R}^n} |\nabla u| \, dy \right)^{\frac{n}{n-1}}. \end{aligned} \tag{3.3}$$

This gives the estimate for $p = 1$ with $p^* = \frac{n}{n-1}$.

Step 2: The case $1 < p < n$. Let $\gamma > 1$ be a constant to be fixed later, and define $v = |u|^\gamma$. Applying (3.3) from step 1 to v , we get

$$\left(\int_{\mathbb{R}^n} |u|^{\gamma \cdot \frac{n}{n-1}} \, dx \right)^{\frac{n-1}{n}} \leq \int_{\mathbb{R}^n} |\nabla |u|^\gamma| \, dx = \gamma \int_{\mathbb{R}^n} |u|^{\gamma-1} |\nabla u| \, dx.$$

Apply Hölder's inequality to the right-hand side:

$$\gamma \int_{\mathbb{R}^n} |u|^{\gamma-1} |\nabla u| \, dx \leq \gamma \left(\int_{\mathbb{R}^n} |u|^{(\gamma-1) \cdot \frac{p}{p-1}} \, dx \right)^{\frac{p-1}{p}} \left(\int_{\mathbb{R}^n} |\nabla u|^p \, dx \right)^{\frac{1}{p}}.$$

Now choose γ such that

$$\gamma \cdot \frac{n}{n-1} = (\gamma-1) \cdot \frac{p}{p-1}.$$

Solving this equation for γ yields

$$\gamma = \frac{p(n-1)}{n-p},$$

so that

$$\gamma \cdot \frac{n}{n-1} = \frac{np}{n-p} = p^*.$$

Substituting, we obtain

$$\left(\int_{\mathbb{R}^n} |u|^{p^*} \, dx \right)^{\frac{1}{p^*}} \leq C \left(\int_{\mathbb{R}^n} |\nabla u|^p \, dx \right)^{\frac{1}{p}}. \quad \clubsuit$$

Theorem 3.4 (Morrey inequality). *Let $n < p \leq +\infty$. Then*

$$\|u\|_{C^{0,1-\frac{n}{p}}(\mathbb{R}^n)} \lesssim_{n,p} \|u\|_{W^{1,p}(\mathbb{R}^n)}.$$

Proof. We claim there exists a constant C , depending only on n , such that

$$\oint_{B_r(x)} |u(y) - u(x)| \, dy \leq C \int_{B_r(x)} \frac{|\nabla u(y)|}{|y - x|^{n-1}} \, dy. \quad (3.4)$$

for each ball $B_r(x) \subseteq \mathbb{R}^n$.

To prove this, fix any point $w \in \partial B_1(0)$. Then if $0 < s < r$,

$$\begin{aligned} |u(x + sw) - u(x)| &= \left| \int_0^s \frac{d}{dt} u(x + tw) \, dt \right| = \left| \int_0^s \nabla u(x + tw) \cdot w \, dt \right| \\ &\leq \int_0^s |\nabla u(x + tw)| \, dt. \end{aligned}$$

Hence

$$\int_{\partial B_1(0)} |u(x + sw) - u(x)| \, d\mathcal{H}^{n-1}(w) \leq \int_0^s \int_{\partial B_1(0)} |\nabla u(x + tw)| \, d\mathcal{H}^{n-1}(w) \, dt. \quad (3.5)$$

Now

$$\begin{aligned} \int_0^s \int_{\partial B_1(0)} |\nabla u(x + tw)| \, d\mathcal{H}_w^{n-1} \, dt &= \int_0^s \int_{\partial B_1(0)} \frac{|\nabla u(y)|}{t^{n-1}} \, d\mathcal{H}_w^{n-1} \, dt \\ &= \int_{B_s(x)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} \, dy \\ &\leq \int_{B_r(x)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} \, dy, \end{aligned}$$

where we put $y = x + tw$, $t = |x - y|$. Furthermore

$$\int_{\partial B_1(0)} |u(x + sw) - u(x)| \, d\mathcal{H}_w^{n-1} = \frac{1}{s^{n-1}} \int_{B_s(x)} |u(z) - u(x)| \, d\mathcal{H}_z^{n-1},$$

for $z = x + sw$. Using the preceding two calculations in (3.5), we obtain the estimate

$$\int_{\partial B_s(x)} |u(z) - u(x)| \, d\mathcal{H}_z^{n-1} \leq s^{n-1} \int_{B_r(x)} \frac{|\nabla u(y)|}{|x - y|^{n-1}} \, dy.$$

This proves our claim (3.4).

We now fix $x \in \mathbb{R}^n$. We apply (3.4) as follows:

$$|u(x)| \leq \oint_{B_1(x)} |u(x) - u(y)| \, dy + \oint_{B_1(x)} |u(y)| \, dy$$

$$\begin{aligned}
&\lesssim_n \int_{B_1(x)} \frac{|\nabla u(y)|}{|x-y|^{n-1}} dy + \|u\|_{L^p(B_1(x))} \\
&\lesssim_n \left(\int_{\mathbb{R}} |\nabla u|^p dz \right)^{1/p} \left(\int_{B_1(x)} \frac{1}{|x-y|^{(n-1)\frac{p}{p-1}}} dy \right)^{\frac{p-1}{p}} + \|u\|_{L^p(\mathbb{R}^n)} \\
&\lesssim_n \|u\|_{W^{1,p}(\mathbb{R}^n)}.
\end{aligned}$$

The last estimate holds since $p > n$ implies $(n-1)\frac{p}{p-1} < n$ so that

$$\int_{B_1(x)} \frac{1}{|x-y|^{(n-1)\frac{p}{p-1}}} dy < +\infty$$

As $x \in \mathbb{R}^n$ is arbitrary, it follows that

$$\sup_{\mathbb{R}^n} |u| \lesssim_n \|u\|_{W^{1,p}(\mathbb{R}^n)}. \quad (3.6)$$

Next, choose any two points $x, y \in \mathbb{R}^n$ and write $r := |x-y|$. Let $W := B_r(x) \cap B_r(y)$. Then

$$|u(x) - u(y)| \leq \int_W |u(x) - u(z)| dz + \int_W |u(y) - u(z)| dz. \quad (3.7)$$

But (3.4) allows us to estimate

$$\begin{aligned}
\int_W |u(x) - u(z)| dz &\lesssim_n \int_{B_r(x)} |u(x) - u(z)| dz \\
&\lesssim_n \left(\int_{B_r(x)} |\nabla u|^p dz \right)^{1/p} \left(\int_{B_r(x)} \frac{dz}{|x-z|^{(n-1)\frac{p}{p-1}}} \right)^{\frac{p-1}{p}} \\
&\lesssim_n \left(r^{n-(n-1)\frac{p-1}{p}} \right)^{\frac{p-1}{p}} \|\nabla u\|_{L^p(\mathbb{R}^n)} \\
&= Cr^{1-\frac{n}{p}} \|\nabla u\|_{L^p(\mathbb{R}^n)}
\end{aligned} \quad (3.8)$$

Likewise,

$$\int_W |u(y) - u(z)| dz \lesssim_n r^{1-\frac{n}{p}} \|\nabla u\|_{L^p(\mathbb{R}^n)}.$$

substituting this estimate and (3.8) into (3.7) yields

$$|u(x) - u(y)| \lesssim_n r^{1-\frac{n}{p}} \|\nabla u\|_{L^p(\mathbb{R}^n)} = |x-y|^{1-\frac{n}{p}} \|\nabla u\|_{L^p(\mathbb{R}^n)}.$$

Thus

$$[u]_{C^{0,1-n/p}(\mathbb{R}^n)} = \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x-y|^{1-n/p}} \lesssim_n \|\nabla u\|_{L^p(\mathbb{R}^n)}.$$

♣

Theorem 3.5 (Approximation). *Let $u \in W^{1,p}(\Omega)$ for $p \in [1, +\infty)$. Then:*

(1) $u^\varepsilon := \eta_\varepsilon \star u \rightarrow u$ in $W_{\text{loc}}^{1,p}(\Omega)$,

(2) *There exists a sequence $u_m \in C^\infty(\Omega)$ such that*

$$u_m \xrightarrow{m \rightarrow +\infty} u \text{ in } W^{1,p}(\Omega).$$

(3) *If $\partial\Omega$ is C^1 then there exists $u_m \in C^\infty(\bar{\Omega})$ such that*

$$u_m \xrightarrow{m \rightarrow +\infty} u \text{ in } W^{1,p}(\Omega).$$

Sketch of the proof. (1) This follows by the properties of the mollification and is left to leave as an exercise to the reader.

(2) Exhaust Ω by compact sets $K_i \Subset \Omega$. Choose a partition of unity ζ_i subordinate to K_i , so that

$$u = \sum_{i \geq 1} u \zeta_i.$$

Given $\delta > 0$, by (1), $\|u \zeta_i - u^i\|_{W^{1,p}(\Omega)} \leq \frac{\delta}{2^i}$ so we have

$$\left\| u - \sum_i u^i \right\|_{W^{1,p}(\Omega)} \leq \delta, \quad \text{and} \quad \sum_i u^i \in C^\infty(\Omega).$$

By the arbitrariness of $\delta > 0$ we conclude.

(3) By covering (and partition of unity), and suffices to work in a ball near $\partial\Omega$. $\partial\Omega$ is locally a C^1 -graph. “Move up and mollify”:

$$x^\varepsilon = x + \lambda \varepsilon e_n, \quad u_\varepsilon(x) := u(x^\varepsilon), \quad v^\varepsilon := \eta_\varepsilon \star u_\varepsilon \in C^\infty(\bar{V}).$$

Then we have

$$\|\nabla v^\varepsilon - \nabla u\|_{L^p(V)} \leq \|\nabla v^\varepsilon - \nabla u^\varepsilon\|_{L^p(V)} + \|\nabla u^\varepsilon - \nabla u\|_{L^p(V)} \xrightarrow{\varepsilon \searrow 0} 0$$

where $\|\nabla v^\varepsilon - \nabla u^\varepsilon\|_{L^p(V)} \rightarrow 0$ by properties of the mollification and $\|\nabla u^\varepsilon - \nabla u\|_{L^p(V)} \rightarrow 0$ by the continuity of the translation in L^p . For details see [2]. ♣

Theorem 3.6 (Extension). *Let Ω be a C^1 bounded domain and suppose $\Omega \Subset K$ in some compact set. Then there exists*

$$E : W^{1,p}(\Omega) \rightarrow W^{1,p}(\mathbb{R}^n)$$

such that

$$Eu \Big|_{\Omega} = u, \text{supp}(Eu) \subseteq K \text{ and } \|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C \|u\|_{W^{1,p}(\Omega)}.$$

Theorem 3.7 (Compactness, Rellich-Kondrachov). *Let $\Omega \subseteq \mathbb{R}^n$ be bounded, open and suppose $\partial\Omega$ to be C^1 . Let $p \in [1, n)$. Then*

$$W^{1,p}(\Omega) \Subset L^q(\Omega), \quad \text{for any } q \in [1, p^*)$$

i.e., for any sequence $u_m \in W^{1,p}(\Omega)$ there is a convergent subsequence in $L^q(\Omega)$

Remark 3.8. The embedding $W^{1,p}(\Omega) \subseteq L^q(\Omega)$ is continuous for any $q \in [1, p^*]$. ♦

Theorem 3.9. *Let $f \in H^{-1}(\Omega) = (H_0^1(\Omega))^*$ and let $a_{i,j}, b_i, c \in L^\infty(\Omega)$ for $i, j = 1, \dots, n$ such that*

$$\sum_{i,j=1}^n a_{i,j} \xi_i \xi_j \geq \lambda |\xi|^2, \quad \text{for any } \xi \in \mathbb{R}^n \quad (3.9)$$

Then the problem

$$\underbrace{\int_{\Omega} \sum_{i,j=1}^n a_{i,j} a_{i,j} \frac{\partial u}{\partial x_i} \frac{\partial v}{\partial x_j} + \sum_{i=1}^n b_i \frac{\partial u}{\partial x_i} v + cuv \, dx}_{=B[u,v]} = \underbrace{\int_{\Omega} f v \, dx}_{=\langle f, v \rangle_{L^2}} \quad \text{for any } v \in H_0^1(\Omega)$$

has a unique solution $u \in H_0^1(\Omega)$.

4 Regularity estimates for the Poisson equation

The goal of this chapter is to make regularity estimates for functions u solving

$$-\Delta u = f$$

using the regularity of f . We will see that

- Schauder: $f \in C^\alpha \implies u \in C^{2,\alpha}$ for $\alpha \in (0, 1)$, and
- Calderón-Zygmund: $f \in L^p \implies W^{2,p}$, $p \in (1, +\infty)$

(of course only if all the requirements for those theorems are met). The $C^{k,\alpha}$ norm is given by

$$\|u\|_{C^{2,\alpha}} = \|u\|_{C^2} + [u]_{C^\alpha}, \quad \text{where} \quad [u]_{C^\alpha} = \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha},$$

where the supremum is over the space we are currently in. We will prove the Schauder estimate by potential theory.

Proposition 4.1. *Let Φ be the fundamental solution of the Laplacian operator, that is*

$$\Phi(x) := \begin{cases} c_n |x|^{2-n}, & n \geq 3 \\ c_2 \log \frac{1}{|x|}, & n = 2, \end{cases}$$

and $f \in C_c^\alpha(\mathbb{R}^n)$, for some $\alpha \in (0, 1]$. Set $v := \Phi \star f$. Then, for all $i, j \in \{1, \dots, n\}$,

$$\partial_{i,j} v(x) = \int_{\mathbb{R}^n} \partial_{i,j} \Phi(x - y) (f(x) - f(y)) \, dy + \frac{\delta_{i,j}}{n} f(x).$$

Here and in what follows, improper and singular integrals are taken in the principal value sense by averaging outside the singularity: for instance,

$$\int_{\mathbb{R}^n} \partial_{i,j} \Phi(x - y) (f(x) - f(y)) \, dy := \lim_{R \rightarrow +\infty} \int_{B_R(x)} \int_{\mathbb{R}^n} \partial_{i,j} \Phi(x - y) (f(x) - f(y)) \, dy.$$

Note that $\partial_{i,j} \Phi$ is indeed not integrable, and hence the integral on the left-hand side of the previous equation is not a standard Lebesgue integral and instead requires the limit definition introduced on the right-hand side.

Proof. We approximate Φ by

$$\Phi_\varepsilon := \begin{cases} \Phi(\varepsilon) + \frac{\varepsilon^2 - |x|^2}{2n|B_\varepsilon|}, & x \in B_\varepsilon(0) \\ \Phi(x) & \text{otherwise} \end{cases}.$$

Then we have, using that $\Phi_\varepsilon = \Phi$ outside $B_\varepsilon(0)$,

$$\begin{aligned} \int_{\mathbb{R}^n} \partial_{i,j} \Phi_\varepsilon(x - y) \underbrace{f(y)}_{=f(y)-f(x)+f(x)} \, dy &= \int_{\mathbb{R}^n} \partial_{i,j} \Phi(x - y) (f(y) - f(x)) \, dy + \frac{\delta_{i,j}}{n} f(x) \\ &= \int_{\mathbb{R}^n} \partial_{i,j} \Phi(x - y) (f(y) - f(x)) \, dy + \frac{\delta_{i,j}}{n} f(x) \end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{R}^n} \partial_{i,j} \Phi_\varepsilon(x-y) dy f(x) + \int_{B_\varepsilon(x)} \underbrace{\partial_{i,j}(\partial_{i,j} \Phi_\varepsilon - \partial_{i,j} \Phi)(x-y)}_{\sim |x-y|^{-n}} \underbrace{(f(x) - f(y))}_{\lesssim_f |x-y|^\alpha} dy + \frac{\delta_{i,j}}{n} f(x) \\
&= \int_{\mathbb{R}^n} \partial_{i,j} \Phi_\varepsilon(x-y) dy f(x) + O(\varepsilon^\alpha) + \frac{\delta_{i,j}}{n} f(x).
\end{aligned}$$

Using the above together with

$$\int_{B_\varepsilon(x)} \partial_{i,j} \Phi(x-y) dy = \int_{B_\varepsilon} \partial_{i,j} \Phi_\varepsilon(z) dz = \int_{B_\varepsilon} -\frac{\partial_{i,j} |z|^2}{2n} dz = -\frac{\delta_{i,j}}{n}$$

and

$$\int_{B_\varepsilon(x)^c} \partial_{i,j} \Phi(x-y) dy = \begin{cases} \lim_{R \rightarrow +\infty} C_n \int_{B_R(x) \setminus B_\varepsilon(x)} \frac{(x_i - y_i)(x_j - y_j)}{|x-y|^{n+2}} = 0, & i \neq j, \\ \lim_{R \rightarrow +\infty} \frac{1}{n} \int_{B_R(x) \setminus B_\varepsilon(x)} \Delta \Phi(x-y) dy = 0, & i = j, \end{cases}$$

where we used odd symmetry if $i \neq j$ and that Φ is harmonic outside its singularity, we deduce that

$$\int_{\mathbb{R}^n} \partial_{i,j} \Phi_\varepsilon(x-y) f(y) dy \xrightarrow{\varepsilon \searrow 0} \int_{\mathbb{R}^n} \partial_{i,j} \Phi(x-y) f(x) - f(y) dy + \frac{\delta_{i,j}}{n} f(x).$$

Also,

$$\begin{aligned}
|\partial_i(\Phi_\varepsilon \star f)(x) - \partial_i(\Phi \star f)(x)| &= |(\partial_i \Phi_\varepsilon - \partial_i \Phi) \star f(x)| \\
&\leq \int_{B_\varepsilon} (|\partial_i \Phi_\varepsilon(z)| + |\partial_i \varphi(z)|) \|f\|_{L^\infty(\mathbb{R}^n)} dz \\
&\leq \int_{B_\varepsilon} \left(\frac{|z|}{\varepsilon^n} + |z|^{1-n} \right) dz \|f\|_{L^\infty(\mathbb{R}^n)} \\
&\lesssim_n \varepsilon \|f\|_{L^\infty(\mathbb{R}^n)}.
\end{aligned}$$

Hence we have

$$\begin{cases} \partial_j(\partial_j \Phi_\varepsilon \star f) \rightarrow \int_{\mathbb{R}^n} \partial_{i,j} \Phi(x-y) f(x) - f(y) dy + \frac{\delta_{i,j}}{n} f(x) \\ \partial_i \Phi_\varepsilon \star f \rightarrow \partial_i \Phi \star f \end{cases}$$

which proves our formula. ♣

In view of the above proposition it is now convenient to study the property of a singular integral kernel $K : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$ satisfying the following properties:

- There exists $g \in L^\infty(\partial B_1)$ such that for any $x \in \mathbb{R}^n \setminus \{0\}$ we have $K(x) = \frac{1}{|x|^n} g\left(\frac{x}{|x|}\right)$ and
- there exists $C > 0$ such that for any $x \in \mathbb{R}^n \setminus \{0\}$ we have that $|\nabla K(x)| \leq \frac{C}{|x|^{n+1}}$.

In short, given an open set $V \subseteq \mathbb{R}^n$ and a function $\Phi \in L^1_{\text{loc}}(\mathbb{R}^n \times \mathbb{R}^n)$ we also use the principal value notation

$$\int_V K(x-y)\Phi(x,y) \, dy := \lim_{\substack{R \rightarrow +\infty \\ \varepsilon \searrow 0}} \int_{(V \setminus B_\varepsilon(x)) \cap B_R(x)} K(x-y)\Phi(x,y) \, dy,$$

whenever the above limit exists.

We also define the integral operator

$$Tf(x) := \int_V K(x-y)(f(x) - f(y)) \, dy.$$

In short, given an open set $V \subset \mathbb{R}^n$ and a function $\Phi \in L^1_{\text{loc}}(\mathbb{R}^n \times \mathbb{R}^n)$, we also use the principal value notation

$$\int_V K(x-y)\Phi(x,y) \, dy := \lim_{R \rightarrow +\infty, \varepsilon \searrow 0} \int_{(V \setminus B_\varepsilon(x)) \cap B_R(x)} K(x-y)\Phi(x,y) \, dy, \quad (4.1)$$

whenever the above limits exist. We also set

$$Tf(x) := \int_V K(x-y)(f(x) - f(y)) \, dy. \quad (4.2)$$

Lemma 4.2. *Assume that T is the operator from above with K as above. Suppose also that for each $z_0 \in \mathbb{R}^n$ and $\rho > 0$,*

$$\left| \int_{V \setminus B_\rho(z_0)} K(z_0 - y) \, dy \right| \leq C_0 \quad (4.3)$$

for some suitable $C_0 > 0$. If $f \in C^\alpha(V)$ for some $\alpha \in (0, 1)$, then

$$[Tf]_{C^\alpha(V)} \lesssim_{n,\alpha,K,C_0} [f]_{C^\alpha(V)}.$$

Proof. Let $z_0, z_1 \in V$. Let $\delta := |z_0 - z_1|$ and $z_2 := \frac{z_0 + z_1}{2}$. The gist of the following estimate is that in regions of order δ around z_2 , all contributions are of order δ^α , and for the complement of these regions, the computation becomes more delicate, and further cancellations must be taken into account. More precisely, we have

$$\begin{aligned} & |Tf(z_0) - Tf(z_1)| \\ &= \left| \int_V K(z_0 - y)(f(z_0) - f(y)) \, dy - \int_V K(z_1 - y)(f(z_1) - f(y)) \, dy \right| \\ &\leq \left| \int_{V \setminus B_{2\delta}(z_2)} K(z_0 - y)(f(z_0) - f(y)) \, dy - \int_{V \setminus B_{2\delta}(z_2)} K(z_1 - y)(f(z_1) - f(y)) \, dy \right| \\ &\quad + \left| \int_{V \cap B_{2\delta}(z_2)} K(z_0 - y)(f(z_0) - f(y)) \, dy \right| \end{aligned}$$

$$+ \left| \int_{V \cap B_{2\delta}(z_2)} K(z_1 - y) (f(z_1) - f(y)) dy \right|. \quad (4.4)$$

Now, for $j \in \{0, 1\}$, if $y \in V \cap B_{2\delta}(z_2)$, then $|z_j - y| \leq |z_j - z_2| + |z_2 - y| < 3\delta$, whence

$$\begin{aligned} & \left| \int_{V \cap B_{2\delta}(z_2)} K(z_j - y) (f(z_j) - f(y)) dy \right| \\ & \leq [f]_{C^\alpha(V)} \int_{B_{2\delta}(z_2)} |K(z_j - y)| |z_j - y|^\alpha dy \\ & \leq [f]_{C^\alpha(V)} \|g\|_{L^\infty(\partial B_1)} \int_{B_{2\delta}(z_2)} |z_j - y|^{\alpha-n} dy \\ & \leq [f]_{C^\alpha(V)} \|g\|_{L^\infty(\partial B_1)} \int_{B_{3\delta}} |x|^{\alpha-n} dx \\ & \leq C[f]_{C^\alpha(V)} \|g\|_{L^\infty(\partial B_1)} \delta^\alpha, \end{aligned} \quad (4.5)$$

for some $C > 0$, thanks to the properties of the kernel K .

Furthermore, making use of fact that the gradient of K is bounded by $\frac{C}{|x|^{n+1}}$, if $y \in V \setminus B_{2\delta}(z_2)$ and $t \in [0, 1]$, then

$$\begin{aligned} |z_1 + t(z_0 - z_1) - y| & \geq |y - z_2| - |z_2 - z_1| - |z_0 - z_1| \\ & \geq |y - z_2| - \frac{3\delta}{2} \geq \frac{3}{4}|y - z_2|, \end{aligned}$$

from which, we deduce (after renaming C if necessary),

$$\begin{aligned} |K(z_0 - y) - K(z_1 - y)| & \leq \int_0^1 |\nabla K(z_1 + t(z_0 - z_1) - y)| dt \cdot |z_0 - z_1| \\ & \leq \sup_{t \in [0,1]} \frac{C|z_0 - z_1|}{|z_1 + t(z_0 - z_1) - y|^{n+1}} \\ & \leq \frac{C\delta}{|y - z_2|^{n+1}}. \end{aligned}$$

For this reason,

$$\begin{aligned} \Xi & := \left| \int_{V \setminus B_{2\delta}(z_2)} K(z_0 - y) (f(z_0) - f(y)) dy - \int_{V \setminus B_{2\delta}(z_2)} K(z_1 - y) (f(z_1) - f(y)) dy \right| \\ & \leq \left| \int_{V \setminus B_{2\delta}(z_2)} K(z_0 - y) (f(z_0) - f(y)) dy - \int_{V \setminus B_{2\delta}(z_2)} K(z_0 - y) (f(z_1) - f(y)) dy \right| \\ & \quad + \left| \int_{V \setminus B_{2\delta}(z_2)} K(z_0 - y) (f(z_1) - f(y)) dy - \int_{V \setminus B_{2\delta}(z_2)} K(z_1 - y) (f(z_1) - f(y)) dy \right| \\ & \leq \left| \int_{V \setminus B_{2\delta}(z_2)} K(z_0 - y) (f(z_0) - f(z_1)) dy \right| + C\delta \int_{V \setminus B_{2\delta}(z_2)} \frac{|f(z_1) - f(y)|}{|y - z_2|^{n+1}} dy. \end{aligned}$$

$$\begin{aligned} &\leq |f(z_0) - f(z_1)| \cdot \left| \int_{B_{8\delta}(z_0) \setminus B_{2\delta}(z_2)} K(z_0 - y) dy \right| + \left| \int_{V \setminus B_{8\delta}(z_0)} K(z_0 - y) dy \right| \\ &\quad + C[f]_{C^\alpha(V)} \delta \int_{\mathbb{R}^n \setminus B_{2\delta}(z_2)} \frac{|z_1 - y|^\alpha}{|y - z_2|^{n+1}} dy. \end{aligned}$$

We also point out that, in view of $K(x) = \frac{1}{|x|^n} g\left(\frac{x}{|x|}\right)$,

$$\begin{aligned} \int_{B_{8\delta}(z_0) \setminus B_{2\delta}(z_2)} |K(z_0 - y)| dy &\leq \|g\|_{L^\infty(\partial B_1)} \int_{B_{8\delta}(z_0) \setminus B_{\delta/8}(z_0)} \frac{dy}{|z_0 - y|^n} \\ &\leq C \|g\|_{L^\infty(\partial B_1)}. \end{aligned}$$

Recalling (4.3), from the above observations we obtain:

$$\begin{aligned} \Xi &\leq (C \|g\|_{L^\infty(\partial B_1)} + C_0) |f(z_0) - f(z_1)| + C[f]_{C^\alpha(V)} \delta \int_{\mathbb{R}^n \setminus B_{2\delta}(z_2)} \frac{|z_1 - y|^\alpha}{|y - z_2|^{n+1}} dy \\ &\leq (C \|g\|_{L^\infty(\partial B_1)} + C_0) [f]_{C^\alpha(V)} \delta^\alpha + C[f]_{C^\alpha(V)} \delta \int_{\mathbb{R}^n \setminus B_{2\delta}(z_2)} \frac{(|z_1 - z_2| + |y - z_2|)^\alpha}{|y - z_2|^{n+1}} dy \\ &\leq (C \|g\|_{L^\infty(\partial B_1)} + C_0) [f]_{C^\alpha(V)} \delta^\alpha + C[f]_{C^\alpha(V)} \delta \int_{\mathbb{R}^n \setminus B_{2\delta}(z_2)} |y - z_2|^{\alpha-n-1} dy \\ &\leq (C \|g\|_{L^\infty(\partial B_1)} + C_0) [f]_{C^\alpha(V)} \delta^\alpha + C[f]_{C^\alpha(V)} \delta^\alpha. \end{aligned}$$

Combined with (4.4) and (4.5), this yields

$$\begin{aligned} |Tf(z_0) - Tf(z_1)| &\leq \Xi + C[f]_{C^\alpha(V)} \|g\|_{L^\infty(\partial B_1)} \delta^\alpha \\ &\leq C[f]_{C^\alpha(V)} (\|g\|_{L^\infty(\partial B_1)} + 1) \delta^\alpha, \end{aligned}$$

providing the desired result. ♣

Consequently, we have

$$-\partial_{i,j} v(x) = \int_{\mathbb{R}^n} \partial_{i,j} \Phi(x - y) \left(f(x) - f(y) \right) dy + \frac{\delta_{i,j}}{n} f(x).$$

From this we deduce:

Corollary 4.3. *It holds that*

$$[\partial_{i,j} v]_{C^\alpha(\mathbb{R}^n)} \lesssim [f]_{C^\alpha(\mathbb{R}^n)}.$$

Proof. From the above discussion we have $\partial_{i,j} v(x) = Tf(x) + \frac{\delta_{i,j}}{n} f(x)$. For $K = \partial_{i,j} \Phi$ and $V = \mathbb{R}^n$ we obtain by symmetry

$$\int_{\mathbb{R}^n \setminus B_\rho(z)} K(z - y) dy = 0. \quad \clubsuit$$

Lemma 4.4. *It holds that*

$$\|v\|_{C^1(\mathbb{R}^n)} \lesssim_{n, \text{supp } f} \|f\|_{L^\infty(\mathbb{R}^n)}.$$

Proof. We have

$$\begin{aligned} |v(x)| + |\nabla v(x)| &\leq \int_{B_R} \left(\Phi(x-y) + |\nabla \Phi(x-y)| \right) \|f\|_{L^\infty(\mathbb{R}^n)} dy \\ &\lesssim (R^2 + R) \|f\|_{L^\infty(\mathbb{R}^n)}. \end{aligned}$$

♣

Theorem 4.5 (Interior estimates). *If $-\Delta u = f$ in B_1 and $f \in C^\alpha(B_1)$, then*

$$\|u\|_{C^{2,\alpha}(B_{1/2})} \lesssim_{n,\alpha} \|u\|_{L^\infty(B_1)} + \|f\|_{C^\alpha(B_1)}.$$

Proof. Let $\chi \in C_c^\infty(B_{1/8})$ be a smooth cut-off i.e.,

$$0 \leq \chi \leq 1, \quad \chi = 1 \text{ on } B_{3/4} \text{ and } |\nabla \chi| \leq C$$

for some constant $C > 0$. Let $\tilde{f} = f\chi$ and $v = \Phi \star \tilde{f}$. These functions then satisfy

$$-\Delta v = \tilde{f} \quad \text{in } \mathbb{R}^n \quad \text{and} \quad \|v\|_{C^{2,\alpha}(B_1)} \leq C \|\tilde{f}\|_{C^\alpha(\mathbb{R}^n)} \leq C \|f\|_{C^\alpha(\mathbb{R}^n)}.$$

Now let $w = u - v$, so w is harmonic in $B_{3/4}$ and

$$\|w\|_{C^3(B_{1/2})} \lesssim \|w\|_{L^\infty(B_{3/4})} \lesssim \|u\|_{L^\infty(B_{3/4})} + \|v\|_{L^\infty(B_{3/4})}$$

implying

$$\begin{aligned} \|u\|_{C^{2,\alpha}(B_{1/2})} &\leq \|v\|_{C^{2,\alpha}(B_{1/2})} + \|w\|_{C^{2,\alpha}(B_{1/2})} \\ &\lesssim \|u\|_{L^\infty(B_{3/4})} + \|f\|_{C^\alpha(B_1)}. \end{aligned}$$

♣

Theorem 4.6 (boundary estimates). *Let $f \in C^\alpha(B_1^+)$, $g \in C^{2,\alpha}(B_1^0)$, $\alpha \in (0, 1)$ and $u \in C^2(B_1^+) \cap C^0(\overline{B_1^+})$ be a solution of*

$$\begin{cases} -\Delta u = f & \text{in } B_1^+ \\ u = g & \text{on } B_1^0, \end{cases}$$

where $B_1^+ = \mathbb{H}^n \cap B_1$ and $B_1^0 = \partial \mathbb{H}^n \cap B_1$. Then

$$\|u\|_{C^{2,\alpha}(B_{1/2}^+)} \lesssim \|u\|_{L^\infty(B_1^+)} + \|f\|_{C^\alpha(B_1^+)} + \|g\|_{C^{2,\alpha}(B_1^0)}$$

Proof. Without loss of generality let $g \equiv 0$ (by considering $\tilde{u} = u - g$, $\tilde{f} = f - \Delta g$). The Newtonian potential on the half space $\mathbb{H}^n$

$$v(x) = \int_{\mathbb{H}^n} \Phi(x - y) f(y) \, dy, \quad \text{supp}(f) \subseteq B_{15/16}^+$$

satisfies

$$\partial_{i,j} v(x) = \int_{\mathbb{H}^n} \partial_{i,j} \Phi(x - y) \left(f(y) - f(x) \right) \, dy + \frac{\delta_{i,j}}{2n} f(x)$$

for any $r > 0$ and for all $x \in B_r^+$ and $(i, j) \neq (n, n)$. For $(i, j) = (n, n)$ use $-\Delta v = f$ to write

$$\partial_{n,n} v = - \sum_{i=1}^{n-1} \partial_{i,i} v - f \quad \text{in } \mathbb{H}^n.$$

Similar arguments lead to

$$[\partial_{i,j} v]_{C^\alpha(\mathbb{H}^n)} \lesssim [f]_{C^\alpha(\mathbb{H}^n)}, \quad \text{and} \quad \|\partial_{i,j} v\|_{L^\infty(\mathbb{H}^n)} \lesssim \|f\|_{C^\alpha(\mathbb{H}^n)}.$$

It suffices to consider a solution to

$$\begin{cases} -\Delta w = f & \text{in } \mathbb{H}^n \\ w = 0 & \text{on } \partial\mathbb{H}^n, \end{cases}$$

i.e., using Green's function for the half plane,

$$w(x) = \int_{\mathbb{H}^n} G_{\mathbb{H}^n}(x, y) f(y) \, dy = \int_{\mathbb{H}^n} (\Phi(x - y) - \Phi(x - y^*)) f(y) \, dy.$$

When subtracting from u we obtain

$$\begin{cases} -\Delta(u - w) = 0 & \text{in } B_1^+, \\ u - w = 0 & \text{on } B_1^0. \end{cases}$$

Reflecting oddly to B_1 gives

$$\|u - w\|_{C^3(B_{1/2})} \lesssim \|u - w\|_{L^\infty(B_1^+)}.$$

Using the above potential estimates for w , we are done. ♣

5 Calderón-Zygmund estimates

In this section we prove the so-called Calderón-Zygmund estimate.

Theorem 5.1. *Let $\Omega \in \mathbb{R}^n$ be a bounded domain and $f \in L^p(\Omega)$ for $p \in (1, +\infty)$. Define*

$$w(x) := \Phi \star f(x) = \int_{\Omega} \Phi(x-y)f(y) dy.$$

Then $w \in W^{2,p}(\Omega)$, $-\Delta w = f$ a.e. in Ω and

$$\|D^2 w\|_{L^p(\Omega)} \lesssim \|f\|_{L^p(\Omega)}$$

and moreover

$$\|D^2 w\|_{L^2(\mathbb{R}^n)} = \|f\|_{L^2(\Omega)}$$

We will proceed to prove the theorem with the following steps:

- (1) For $p = 2$ we integrate by parts to obtain the theorem.
- (2) We will prove a weak statement of the theorem $p = 1$ by using a cube decomposition.
- (3) For $1 < p < 2$ we will use Marcinkiewicz interpolation.
- (4) The case $2 < p < +\infty$ will then follow by duality.

5.1 Cube decomposition

We will now start with a cube decomposition. Let $f \in L^1(Q_0)$ where Q_0 is a cube. The goal is to divide f up into “good” and “bad” parts where the “good” parts are in L^∞ and the “bad” ones in $L^1 \setminus L^\infty$ (where spikes can happen).

Let $t \in \mathbb{R}$ such that $\int_{Q_0} f dx \leq t$. Divide Q_0 into 2^n disjoint sub cubes with half the side-lengths of the original cube. Let Q be one of those cubes. If $\int_Q f dx \leq t$ we divide again and if

$$t < \int_Q f dx \leq 2^n t$$

we stop and put Q in the set $\mathcal{S}$ (for spikes). Do this for any all divisions of the original cube Q_0 . Define $\mathcal{G} := Q_0 \setminus \bigcup_{Q \in \mathcal{S}} Q$ (the good set). By Lebesgue’s differentiation theorem we have

$$f(x) = \lim_{\substack{|Q| \rightarrow 0 \\ x \in Q}} \int_Q f dx \leq t, \quad \text{for a.e. } x \in \mathcal{G}.$$

The Volume of cubes in $\mathcal{S}$ is controlled by

$$\sum_{Q \in \mathcal{S}} |Q| \leq \frac{1}{t} \sum_{Q \in \mathcal{S}} \int_Q f dx \leq \frac{1}{t} \int_{Q_0} f dx.$$

We just have proved the following lemma:

Lemma 5.2. *Let $f \in L^1(Q_0)$ and $t \geq \int_{Q_0} f \, dx$. There exists a collection of dyadic cubes $\mathcal{S}$ such that*

$$\begin{cases} t < \int_Q f \, dx \leq s^n t & \text{for all } Q \in \mathcal{S}, \\ f \leq t & \text{a.e. in } Q_0 \setminus \bigcup_{Q \in \mathcal{S}} Q, \\ \sum_{Q \in \mathcal{S}} |Q| \leq \frac{1}{t} \|f\|_{L^1(Q_0)}. \end{cases}$$

5.2 The Marcinkiewicz interpolation theorem

The second tool we need is the distribution function f in $\Omega \in \mathbb{R}^n$ which we define by

$$\mu_f(t) := |\{x \in \Omega \mid |f(x)| > t\}|, \quad t > 0.$$

Note that μ_f is a decreasing function.

Lemma 5.3. *Let $p > 0$ and $|f|^p \in L^1(\Omega)$. Then the following holds:*

$$\mu_f(t) \leq \frac{1}{t^p} \int_{\Omega} |f|^p \, dx \quad (\text{Chebistev})$$

$$\int_{\Omega} |f|^p \, dx = p \int_0^{+\infty} t^{p-1} \mu_f(t) \, dt \quad (\text{Layer cake})$$

Proof. First for $p > 0$ we find

$$\begin{aligned} \mu_f(t) &= \frac{1}{t^p} \int_{\Omega} t^p \mathbf{1}_{|f(x)| > t}(x) \, dx \leq \frac{1}{t^p} \int_{\Omega} |f(x)|^p \mathbf{1}_{|f(x)| > t}(x) \, dx \\ &\leq \frac{1}{t^p} \int_{\Omega} |f(x)|^p \, dx = \frac{1}{t^p} \|f\|_{L^p(\Omega)}^p, \end{aligned}$$

which proves the Chebistev inequality. For the Layer cake formula we use change of variables and Fubini to obtain

$$\begin{aligned} \int_{\Omega} |f|^p \, dx &= \int_{x \in \Omega} \int_{t=0}^{+\infty} \mathbf{1}_{|f(x)| > t^{1/p}} \, dt \, dx \\ &\stackrel{\text{Change of Variables}}{=} p \int_{x \in \Omega} \int_{t=0}^{+\infty} t^{p-1} \mathbf{1}_{|f(x)| > t} \, dt \, dx \\ &\stackrel{\text{Fubini}}{=} p \int_0^{+\infty} t^{p-1} \mu_f(t) \, dt. \end{aligned}$$

♣

Remark 5.4 (Weak L^p -space). If $f \in L^p(\Omega)$ for $p \in [1, +\infty)$ we have that

$$\|f\|_{L^{p,\infty}(\Omega)} := \sup_{t>0} t \mu_f^{1/p}(t) \leq \|f\|_{L^p(\Omega)}$$

is a quasi norm meaning the triangle equality holds up to a constant. For an operator T we say that “ T is of strong type (p, p) ” if

$$\|Tf\|_{L^p} \lesssim \|f\|_{L^p}$$

which implies that “ T is of weak type (p, p) ” i.e.,

$$\|Tf\|_{L^{p,\infty}} \lesssim \|f\|_{L^p}.$$



Theorem 5.5. *Let $1 \leq q < r < +\infty$ and let $T : L^q \cap L^r(\Omega) \rightarrow L^q \cap L^r(\Omega)$ be a linear operator satisfying*

$$\|Tf\|_{L^{q,\infty}(\Omega)} \leq T_1 \|f\|_{L^q(\Omega)} \quad (\text{Weak type } (q, q))$$

$$\|Tf\|_{L^{r,\infty}(\Omega)} \leq T_2 \|f\|_{L^r(\Omega)} \quad (\text{Weak type } (r, r))$$

then for any p with $q < p < r$ and $p = \frac{\theta}{q} + \frac{1-\theta}{r}$ for $\theta \in (0, 1)$, T is a map $T : L^p(\Omega) \rightarrow L^p(\Omega)$ and satisfies the estimate

$$\|Tf\|_{L^p(\Omega)} \lesssim_{p,q,r} T_1^\theta T_2^{1-\theta} \|f\|_{L^p(\Omega)}. \quad (\text{Strong type } (p, p))$$

Proof. For $f \in L^q \cap L^r(\Omega)$ and $s > 0$ we write $f = f_1 + f_2$ where

$$f_1(x) := \begin{cases} f(x), & \text{if } |f(x)| > s, \\ 0, & \text{if } |f(x)| \leq s, \end{cases} \quad f_2(x) = \begin{cases} 0, & \text{if } |f(x)| > s, \\ f(x), & \text{if } |f(x)| \leq s. \end{cases}$$

Then $|Tf| \leq |Tf_1| + |Tf_2|$ and hence

$$\begin{aligned} \mu_{Tf} &\leq \mu_{Tf_1} \left(\frac{t}{2} \right) + \mu_{Tf_2} \left(\frac{t}{2} \right) \\ &\leq \left(\frac{2T_1}{t} \right)^q \int_{\Omega} |f_1|^q dx + \left(\frac{2T_2}{t} \right)^r \int_{\Omega} |f_2|^r dx. \end{aligned}$$

Using Lemma 5.3 we obtain

$$\begin{aligned} \int_{\Omega} |Tf|^p dx &= p \int_0^{+\infty} t^{p-1} \mu_{Tf}(t) dt \\ &\leq p(2T_1)^q \int_0^{+\infty} t^{p-1-q} \left(\int_{|f|>s} |f|^q dx \right) dt \\ &\quad + p(2T_2)^r \int_0^{+\infty} t^{p-1-r} \left(\int_{|f|\leq s} |f|^r dx \right) dt. \end{aligned}$$

We now choose s as a function of t ; in particular we take $t = As$ for some $A > 0$ to be fixed later. Then, we have

$$\begin{aligned} \int_{\Omega} |Tf|^p dx &\leq p(2T_1)^q A^{p-q} \int_0^{+\infty} s^{p-1-q} \int_{|f|<s} |f|^q dx ds \\ &\quad + p(2T_2)^r A^{p-r} \int_0^{+\infty} s^{p-1-r} \int_{|f|\leq s} |f|^r dx ds \end{aligned}$$

But

$$\begin{aligned} \int_0^{+\infty} s^{p-1-q} \int_{|f|>s} |f|^q dx ds &= \int_{\Omega} |f(x)|^q \int_0^{|f(x)|} s^{p-1-q} ds dx \\ &= \frac{1}{p-q} \int_{\Omega} |f|^p dx, \end{aligned}$$

and, similarly,

$$\begin{aligned} \int_0^{+\infty} s^{p-1-r} \int_{|f|\leq s} |f|^r dx ds &= \int_{\Omega} |f(x)|^r \int_{|f(x)|}^{+\infty} s^{p-1-r} ds dx \\ &= \frac{1}{r-p} \int_{\Omega} |f|^p dx. \end{aligned}$$

Consequently,

$$\int_{\Omega} |Tf|^p dx \leq \left\{ \frac{p}{p-q} (2T_1)^q A^{p-q} + \frac{p}{r-p} (2T_2)^r A^{p-r} \right\} \int_{\Omega} |f|^p dx$$

for any $A > 0$. This means we can just take the number of A for which the expression in the braces is minimal. By taking the derivate we find

$$A = 2T_1^{\frac{q}{r-q}} T_2^{\frac{r}{p-q}}$$

we thus obtain

$$\|Tf\|_{L^p(\Omega)} \leq 2 \left(\frac{p}{p-q} + \frac{p}{r-p} \right)^{1/p} T_1^{\alpha} T_2^{1-\alpha} \|f\|_{L^p(\Omega)}.$$

This implies the statement of the theorem. ♣

5.3 Proof of Theorem 5.1

Proof. **1.** We first deal with the case $p = 2$. Let $f \in C_c^\infty(\mathbb{R}^n)$ and take $R > 0$ such that $\text{supp}(f) \subseteq B_R(0) =: B_R$. Recall that $\|w\|_{C^1(\mathbb{R}^n)} \leq \|f\|_{L^\infty(B_R)}$ and for any $x \in \mathbb{R}^n \setminus B_{2R}$ it holds that

$$|\partial_{i,j} w(x)| \leq \int_{B_R} \frac{|f(y)|}{|x-y|^n} dy \leq \frac{R^n \|f\|_{L^\infty(B_R)}}{|x|}$$

Using Green's first identity twice we obtain

$$\begin{aligned}
\int_{B_R} |D^2 w|^2 dx &= \int_{B_R} \sum_{i,j=1}^n (\partial_{i,j} w)^2 dx \\
&= \sum_{i,j} \int_{B_R} \operatorname{div}(\partial_{i,j} w \partial_j w e_i) - \partial_{i,i,j} w_j w dx \\
&= \sum_{i,j} \left[\int_{\partial B_R} \partial_{i,j} w \partial_j w \nu_j d\mathcal{H}^{n-1} - \int_{B_R} \operatorname{div}(\partial_{i,i} w \partial_j w e_j) - \partial_{i,i} w \partial_{j,j} w \right] dx \\
&= O_{R \rightarrow +\infty} \left(\frac{\mathcal{H}^{n-1}(\partial B_R)}{R^n} \right) + \int_{B_R} (\Delta w)^2 dx \xrightarrow{R \rightarrow +\infty} \int_{\mathbb{R}^n} f^2 dx
\end{aligned}$$

For general $f \in L^2(\Omega)$, approximate by $f_m \in C_c^\infty$ such that $f_m \rightarrow f$ in $L^2(\Omega)$. Then $\Phi \star f_m \rightarrow \Phi \star f$ in $W^{2,2}(\Omega)$ (Exercise).

2. Fix i, j and define the linear operator

$$T : L^2(\Omega) \rightarrow L^2(\Omega), \quad Tf = \partial_{i,j} w + \frac{\delta_{i,j}}{n} f$$

In fact the first step gives us that $T : L^2 \rightarrow L^2$. Moreover we deduce (also from the first step) that

$$\mu_{Tf}(t) \leq \left(\frac{\|f\|_{L^2(\Omega)}}{t} \right)^2$$

We now show that

$$\mu_{Tf}(t) \lesssim \frac{\|f\|_{L^1(\Omega)}}{t} \quad (5.1)$$

To this end, extend f by zero outside of Ω and take a cube Q_0 containing Ω , such that for a fixed $t > 0$ we have

$$\int_{Q_0} f dx \leq t |Q_0|.$$

By cube decomposition (Lemma 5.2) we find a sequence of dyadic cubes $\{Q_j\}_{j \in \mathbb{N}}$ such that

$$t < \int_{Q_j} |f| dx \leq 2^n t, \quad |f| \leq t \text{ a.e. in } G := Q_0 \setminus \bigcup_{j=1}^{+\infty} Q_j \quad (5.2)$$

The function f is now split into a “good part” g defined by

$$g(x) = \begin{cases} f(x) & \text{for all } x \in G \\ \int_{Q_i} f(y) dy & \text{for } x \in Q_i \text{ for } i = 1, 2, \dots \end{cases}$$

and a “bad part” $b = f - g$. Clearly,

$$|g| \leq 2^n t \text{ a.e.}, \quad b \Big|_G \equiv 0, \quad \text{and} \quad \int_{Q_\ell} b dx = 0 \quad \text{for } \ell = 1, 2, \dots$$

Using the linearity of T we find with $Tf = Tg + Tb$ that

$$\mu_{Tf}(t) \leq \mu_{Tg}\left(\frac{t}{2}\right) + \mu_{Tb}\left(\frac{t}{2}\right).$$

3. We now estimate Tg :

$$\mu_{Tg}\left(\frac{t}{2}\right) \leq \frac{4}{t^2} \int_{Q_0} |g|^2 dx \leq \frac{2^{n+2}}{t} \int_{Q_0} |g| \leq \frac{2^{n+2}}{t} \int_{\Omega} |f| dx.$$

4. We now estimate Tb . Split $b = \sum_{j \geq 1} b_j$ where $b_j = b \mathbf{1}_{Q_j}$. Let p_j be the center of Q_j . Fix ℓ and a sequence $\{b_{\ell_m}\}_{m \in \mathbb{N}} \subseteq C_c^\infty(Q_\ell)$ such that

$$\int_{Q_\ell} b_{\ell_m} dx = \int_{Q_\ell} b_\ell dx = 0.$$

Then for $x \notin Q_\ell$ we find the formula

$$Tb_{\ell_m}(x) = \int_{Q_\ell} D_{i,j} K(x-y) b_{\ell_m}(y) dy = \int_{Q_\ell} \left\{ D_{i,j} K(x-y) - D_{i,j}(x-p_\ell) \right\} b_{\ell_m}(y) dy.$$

Further, we get

$$|Tb_{\ell_m}(x)| \lesssim_n \text{diam}(K_\ell) (\text{dist}(x, K_\ell))^{-n-1} \int_{Q_\ell} |b_{\ell_m}(y)| dy$$

Letting $B_\ell = B_{\text{diam}(Q_\ell)}(p_\ell)$ we obtain by integration

$$\int_{Q_0 \setminus B_\ell} |Tb_{\ell_m}| dx \lesssim_n \text{diam}(Q_\ell) \int_{|x| \geq \frac{\text{diam}(Q_\ell)}{2}} \frac{dx}{|x|^{n+1}} \lesssim_n \int_{Q_\ell} |b_{\ell_m}| dx$$

Consequently, letting $m \rightarrow +\infty$, writing $F^* := \bigcup_{\ell \in \mathbb{N}} B_\ell$, $G^* = Q_0 \setminus F^*$ and summing over $\ell \in \mathbb{N}$ we get

$$\int_{G^*} |Tb| dx \lesssim_n \int_{G^*} |b| dx \lesssim_n \int_{G^*} |f| dx,$$

so that by Lemma 5.3

$$|\{x \in G^* \mid |Tb| > t/2\}| \lesssim_n \frac{\|f\|_{L^1}}{t}.$$

However by (5.2),

$$|F^*| \lesssim_n |F| \lesssim_n \frac{\|f\|_{L^1}}{t}.$$

which proves (5.1).

5. By our previous steps we are in position to apply the Marcinkiewicz interpolation theorem which gives us the result for $p \in (1, 2)$. If $p > 2$ we extend by duality. For if $f, g \in C_c^\infty(\Omega)$ then

$$\begin{aligned} \int_{\Omega} (Tf)g \, dx &= \int_{\Omega} w D_{i,j} g \, dx \\ &= \iint_{\Omega \times \Omega} K(x-y) f(y) D_{i,j} g(x) \, dx \, dy \\ &= \int_{\Omega} f Tg \, dx \\ &\leq \|f\|_{L^p} \|Tg\|_{L^{p'}} . \end{aligned}$$

Thus we find for $p > 2$ that

$$\|Tf\|_{L^p} = \sup_{\substack{g \in L^p \\ \|g\|_{L^p}=1}} \int_{\Omega} (Tf)g \, dx \lesssim_{n,p'} \|f\|_{L^{p'}}$$

By the usual approximation by test functions we conclude this proof. ♣

6 Pohožaev identity and applications

The main result of this chapter is the following.

Lemma 6.1. *Let $\Omega \subseteq \mathbb{R}^n$ be bounded with C^1 boundary and $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ be a solution to*

$$-\Delta u = f(u) \quad \text{in } \Omega$$

for some $f \in L^\infty_{\text{loc}}(\mathbb{R})$. Write also

$$F(u) := \int_0^u f(t) dt.$$

Then,

$$\begin{aligned} & \int_{\Omega} nF(u) - \frac{n-2}{2}uf(u) dx \\ &= \int_{\partial\Omega} \left(x \cdot \nabla u + \frac{n-2}{2}u \right) \partial_\nu u + \left(F(u) - \frac{|\nabla u|^2}{2}(x \cdot \nu) \right) d\mathcal{H}_x^{n-1}. \end{aligned}$$

Proof. We compute

$$\operatorname{div} \left[\left(x \cdot \nabla u + \frac{n-2}{2}u \right) \nabla u + \left(F(u) - \frac{|\nabla u|^2}{2}x \right) \right].$$

We have

$$\begin{aligned} \operatorname{div} \left((x \cdot \nabla u) \nabla u \right) &= (x \cdot \nabla u) \Delta u + \nabla(x \cdot \nabla u) \cdot \nabla u \\ &= -(x \cdot \nabla u) f(x, u) + \partial_i(x_j \partial_j u) \partial_i u \\ &= -(x \cdot \nabla u) f(x, u) + (\delta_{i,j} \partial_j u + x_j \partial_{i,j} u) \partial_i u \\ &= -(x \cdot \nabla u) f(x, u) + |\nabla u|^2 + \frac{1}{2} x \cdot \nabla |\nabla u|^2, \end{aligned}$$

$$\operatorname{div} \left[\frac{n-2}{2} u \nabla u \right] = \frac{n-2}{2} (u \Delta u + |\nabla u|^2) = -\frac{n-2}{2} u f(u) + \frac{n-2}{2} |\nabla u|^2,$$

$$\operatorname{div}(F(u)x) = (\operatorname{div} x)F(u) + x \cdot \nabla(F(u)) = nF(u) + (x \cdot \nabla u)f(u),$$

and

$$\operatorname{div} \left(-\frac{|\nabla u|^2}{2} x \right) = -\frac{1}{2} (\operatorname{div} x) |\nabla u|^2 - \frac{1}{2} x \cdot \nabla |\nabla u|^2 = -\frac{n}{2} |\nabla u|^2 - \frac{1}{2} x \cdot \nabla |\nabla u|^2.$$

Putting all those pieces together and cancelling out terms we obtain

$$\operatorname{div} \left[\left(x \cdot \nabla u + \frac{n-2}{2} u \right) \nabla u + \left(F(u) - \frac{|\nabla u|^2}{2} \right) x \right] = nF(u) - \frac{n-2}{2} u f(u).$$

Using the divergence theorem we obtain the result. ♣

Corollary 6.2. *Under the same hypothesis as Lemma 6.1, if*

$$u = 0 \quad \text{on } \partial\Omega$$

we find

$$\int_{\Omega} \left[nF(u) - \frac{n-2}{2} u f(u) \right] dx = \int_{\partial\Omega} \frac{|\nabla u|^2}{2} (x \cdot \nu) d\mathcal{H}_x^{n-1}$$

Proof. Using that $F(0) = 0$ the only boundary terms are

$$\int_{\partial\Omega} \left[(x \cdot \nabla u) \partial_{\nu} u - \frac{|\nabla u|^2}{2} (x \cdot \nu) \right] d\mathcal{H}_x^{n-1}.$$

Since $\partial\Omega$ is contained in the 0-level set $\{u = 0\}$,

$$\nu = \pm \frac{\nabla u}{|\nabla u|}.$$

So on $\partial\Omega$ we have

$$(x \cdot \nabla u) \partial_{\nu} u = (x \cdot \nabla u) (\nabla u \cdot \nu) = |\nabla u|^2 (x \cdot \nu) \underbrace{(\nu \cdot \nu)}_{=|\nu|^2=1}. \quad \clubsuit$$

Definition 6.3. We call a domain **star-shaped** with respect to the origin if

$$x \cdot \nu(x) \geq 0 \quad \text{for a.e. } x \in \partial\Omega.$$

This definition can be generalized to the following: We call a domain Ω star-shaped with respect to $x_0 \in \mathbb{R}^n$ if for any $y \in \Omega$ and for any $t \in (0, 1)$,

$$tx + (1-t)y \in \Omega.$$

We leave it as an exercise to the reader to verify this.

Theorem 6.4. *Let $n \geq 3$, $p > \frac{n+2}{n-2}$ and Ω be a bounded, star-shaped domain with*

C^1 -boundary in $\mathbb{R}^n$. Let $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ be a solution to

$$\begin{cases} -\Delta u = |u|^{p-1}u & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Then $u \equiv 0$.

Proof. Use Lemma 6.1 with $f(u) = |u|^{p-1}u$, i.e., $F(u) = \frac{|u|^{p+1}}{p+1}$. Then we find

$$\frac{1}{2} \int_{\partial\Omega} |\nabla u|^2 (x \cdot \nu) d\mathcal{H}_x^{n-1} = \frac{n}{p+1} \int_{\Omega} |u|^{p+1} dx - \frac{n-2}{2} \int_{\Omega} |u|^{p+1} dx$$

Using that $p > \frac{n+2}{n-2}$, $p+1 > \frac{2n}{n-2} = 2^*$ i.e., $\frac{n}{p+1} < \frac{n-2}{2}$ we obtain

$$\underbrace{\left(\frac{n-2}{2} - \frac{n}{p+1} \right)}_{>0} \int_{\Omega} |u|^{p+1} dx + \frac{1}{2} \int_{\partial\Omega} |\nabla u|^2 \underbrace{(x \cdot \nu)}_{\geq 0} d\mathcal{H}_x^{n-1} = 0$$

thus $\int_{\Omega} |u|^{p+1} dx = 0$. ♣

Remark 6.5. Actually $p = \frac{n+2}{n-2}$ is possible if we additionally require $u \geq 0$ and $x \cdot \nu > 0$ for $\mathcal{H}^{n-1}$ a.e. $x \in \partial\Omega$. We then still have $\partial_\nu u = 0$ but

$$\int_{\partial\Omega} \partial_\nu u d\mathcal{H}^{n-1} = \int_{\Omega} \Delta u dx = - \int_{\Omega} u^{\frac{n+2}{n-2}} dx$$

where the RHS would be strictly smaller than 0 if $u \not\equiv 0$. ♦

Now consider the slightly non-trivial equation

$$-\Delta u = 1 \quad \text{in } \Omega.$$

This is related to “torsion”. For what domain can one prescribe **both** Dirichlet and Neumann boundary conditions. It is possible in a ball:

$$u(x) = \frac{1 - |x|^2}{2n}$$

solves

$$\begin{cases} -\Delta u = 1 & \text{in } B_1 \\ u = 0 & \text{on } \partial B_1 \\ \partial_\nu u = -\frac{1}{n} & \text{on } \partial B_1. \end{cases}$$

We will now prove that there are no other such bounded domains where u can satisfy this.

Theorem 6.6 (Serrin, 1971). *Let $\Omega \subseteq \mathbb{R}^n$ be a bounded C^2 -domain and $u \in C^2(\overline{\Omega})$ be a solution to*

$$\begin{cases} -\Delta u = 1, & \text{in } \Omega, \\ u = 0, & \text{on } \partial\Omega, \\ \partial_\nu u = -c, & \text{on } \partial\Omega. \end{cases}$$

for some $c \in \mathbb{R}$. Then Ω is a ball.

Proof. **1.** Observe that

$$|\Omega| = \int_{\Omega} -\Delta u \, dx = \int_{\partial\Omega} -\partial_\nu u \, d\mathcal{H}^{n-1} = c\mathcal{H}^{n-1}(\partial\Omega),$$

so c is uniquely determined in terms of Ω .

2. Using Pohožaev identity with $f(u) = 1$, $F(u) = u$ we obtain

$$\begin{aligned} \frac{n+2}{2} \int_{\Omega} u \, dx &= \int_{\Omega} nF(u) - \frac{n-2}{2} u f(u) \, dx \\ &= \int_{\partial\Omega} \frac{(\partial_\nu u)^2}{2} (x \cdot \nu) \, d\mathcal{H}_x^{n-1} = \frac{c^2}{2} \int_{\Omega} \operatorname{div} x \, dx = \frac{nc^2}{2} |\Omega|. \end{aligned}$$

3. Now, consider the function

$$P(x) = \frac{n}{2} |\nabla u(x)|^2 + u(x).$$

Note that $P = \frac{nc^2}{2}$ on $\partial\Omega$. We Claim that P is subharmonic in Ω that is

$$\Delta P \geq 0 \quad \text{in } \Omega.$$

Using Bochner identity,

$$\begin{aligned} \frac{1}{2} \Delta |\nabla u|^2 &= \frac{1}{2} \partial_{j,j} (\partial_i u \partial_i u) = \partial_j (\partial_i u \partial_{i,j} u) \\ &= \partial_i u \partial_{i,j,j} u + \partial_{i,j} u \partial_{i,j} u = \nabla u \cdot \nabla \Delta u + |D^2 u|^2 \end{aligned}$$

and the equation $-\Delta u = 1$ in Ω ,

$$\Delta P = n |D^2 u|^2 - 1 = n |D^2 u|^2 - (\Delta u)^2 \geq 0$$

where we used the Cauchy-Schwartz inequality for the inequality.

4. Now P is subharmonic. By the strong maximum principle,

$$P(x) \leq \sup_{\partial\Omega} P = \frac{nc^2}{2},$$

and moreover,

$$\text{either } P(x) < \frac{nc^2}{2} \text{ for any } x \in \Omega \text{ or } P(x) = \frac{nc^2}{2} \text{ for any } x \in \Omega.$$

We claim that P is constant in Ω , i.e., that “=” holds. Indeed,

$$\begin{aligned} \int_{\Omega} P(x) \, dx &= \frac{n}{2} \int_{\Omega} |\nabla u|^2 \, dx + \int_{\Omega} u \, dx = \frac{n}{2} \int_{\Omega} -u \Delta u \, dx + \int_{\Omega} u \, dx \\ &= \frac{n+2}{2} \int_{\Omega} u \, dx. \end{aligned}$$

If “<” holds, then by **2.**,

$$\frac{nc^2}{2} |\Omega| > \int_{\Omega} P(x) \, dx = \frac{n+2}{2} \int_{\Omega} u \, dx = \frac{nc^2}{2} |\Omega|$$

which is a contradiction.

5. We now know that $P \equiv \frac{nc^2}{2}$ and that $\Delta P = n|D^2u|^2 - 1 = 0$ i.e., equality holds in the Cauchy-Schwartz inequality:

$$(\Delta u)^2 = \left(\sum_{i=1}^n \partial_{i,i} u \right)^2 \leq \sum_{i=1}^n (\partial_{i,i} u)^2 \sum_{j=1}^n 1^2 \leq n |D^2u|^2,$$

meaning that $D^2u = c_0 I$ where I denotes the identity matrix. Taking trace gives

$$nc_0 = \Delta u = -1 \implies c_0 = -\frac{1}{n}.$$

6. From $\partial_{i,j} u = -\frac{\delta_{i,j}}{n}$ we just integrate:

$$\begin{aligned} \partial_{i,j} \left(u + \frac{|x|^2}{2n} \right) &= 0 \quad \text{for all } 1 \leq i, j \leq n, \\ \implies \nabla \partial_j \left(u + \frac{|x|^2}{2n} \right) &= 0 \quad \text{for all } 1 \leq j \leq n, \\ \implies \partial_j \left(u + \frac{|x|^2}{2n} \right) &= a_j \in \mathbb{R} \quad \text{for all } 1 \leq j \leq n, \\ \implies \partial_j \left(u + \frac{|x|^2}{2n} - a \cdot x \right) &= 0 \in \mathbb{R} \quad \text{for all } 1 \leq j \leq n, \\ \implies \nabla \left(u + \frac{|x|^2}{2n} - a \cdot x \right) &= 0 \\ \implies u + \frac{|x|^2}{2n} - a \cdot x &= b. \end{aligned}$$

Completing the square gives

$$u(x) = \frac{r^2 - |x - x_0|^2}{2n}.$$

This implies

$$\partial\Omega \subseteq \{u = 0\} \subseteq \partial B_r(x_0),$$

in particular, Ω is a ball. This concludes the proof.

♣

7 Lane-Emden equation

In this chapter we study the so-called **Lane-Emden equation** which reads

$$-\Delta u = u^p, \quad \text{in } \Omega.$$

For simplicity we shall only consider the case $u \geq 0$. We have seen non-existence for $p \geq \frac{n+2}{n-2}$ in star-shaped domains.

For $p \in (1, \frac{n+2}{n-2})$, we have seen **existence** and **regularity**. The goal of this section is to prove the following theorem.

Theorem 7.1. *Let $\Omega \subseteq \mathbb{R}^n$ a bounded smooth domain. Let*

$$p \in \begin{cases} (1, \frac{n+2}{n-2}) & \text{for } n \geq 3 \\ (1, +\infty) & \text{for } n = 1, 2 \end{cases}.$$

Then

$$\begin{cases} -\Delta u = |u|^{p-1} u & \text{in } \Omega, \\ u > 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega \end{cases} \quad (7.1)$$

has a solution in $C^2(\Omega) \cap C^0(\bar{\Omega})$.

The proof of this theorem uses variational (energy) methods, namely the constrained direct method of minimization. For more, see literature (e.g., Struwe's book "Variational Methods").

We start with an abstract result.

Theorem 7.2. *Let V be a reflexive Banach space and $M \subseteq V$ a weakly closed subset. Suppose that $E : M \rightarrow (-\infty, +\infty]$ is*

*(E1) **coercive**, i.e., $E(u) \rightarrow +\infty$ as $\|u\|_V \rightarrow +\infty$, $u \in M$;*

*(E2) **sequentially weakly lower semi-continuous**, that is, for any sequence $u_m \rightharpoonup u$ with $u_m \in M$ we have*

$$E(u) \leq \liminf_{m \rightarrow +\infty} E(u_m);$$

Then E is bounded from below in M and attains its infimum in M .

Proof. Let $u_m \in M$ be a minimizing sequence i.e., $E(u_m) \rightarrow \inf_M E < +\infty$. By (E1), the sequence is bounded. Using that V is reflexive, by Eberlein-Šmulian we find that $u_m \rightharpoonup u$ for some $u \in V$. Since M is weakly closed, $u \in M$. By weak lower semi-continuity (E2),

$$-\infty < E(u) \leq \liminf_{m \rightarrow +\infty} E(u_m) \leq \inf_M E$$

thus $E(u) = \inf_M E$. ♣

Note that (7.1) is the Euler Lagrange equation of

$$\tilde{E}(u) = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx - \frac{1}{p+1} \int_{\Omega} |u|^{p+1} dx$$

However $\tilde{E}$ is neither bounded from above nor from below on $H_0^1(\Omega)$ (exercise).

Proof of Theorem 7.1. **1.** We solve the constrained minimization problem for

$$E[u] = \frac{1}{2} \int_{\Omega} |\nabla u|^2 dx$$

on the Hilbert space $H_0^1(\Omega)$ restricted to the set

$$M = \left\{ u \in H_0^1(\Omega) \mid \int_{\Omega} |u|^{p+1} dx = 1 \right\}.$$

2. To this end, we apply Theorem 7.2 so we check the hypotheses of this theorem.

M is weakly closed: If $u_m \in M \subseteq H_0^1(\Omega)$ and $u_m \rightharpoonup u$ weakly in $H_0^1(\Omega)$, by Rellich-Kondrachov compactness theorem, $H_0^1 \hookrightarrow L^{p+1}(\Omega)$ is compact for $p+1 < 2^* = \frac{2n}{n-2}$ for $n \geq 3$ (or any $p \in \mathbb{R}_{\geq 1}$ if $n = 1, 2$). Hence, up to subsequence, $u_m \rightarrow \tilde{u}$ in $L^{p+1}(\Omega)$. We have that $u = \tilde{u}$ since convergence in H_0^1 implies convergence in L^2 and strong convergence in L^2 implies weak convergence in L^2 and weak limits are unique. By the convergence in $p+1$ we have

$$\|u\|_{L^{p+1}(\Omega)} = \lim_{m \rightarrow +\infty} \|u_m\|_{L^{p+1}(\Omega)} = 1, \quad \text{i.e., } u \in M.$$

E is coercive. This is obvious because $E(u) = \|u\|_{H_0^1(\Omega)}^2$ up to possibly a scalar multiple.

E is weakly lower semi-continuous. This follows by the weak lower semi-continuity of the norm.

By Theorem 7.2 $\min_M E$ exists at some $\underline{u} \in M$. Since $E(u) = E(|u|)$ we may without loss of generality assume $\underline{u} \geq 0$.

3. Let

$$G[u] := \int_{\Omega} |u|^{p+1} dx - 1$$

so $M = G^{-1}\{0\}$. The functionals $E, G : H_0^1(\Omega) \rightarrow \mathbb{R}$ is continuously Fréchet-differentiable with

$$\begin{aligned} \langle E'[u], v \rangle &= \int_{\Omega} \nabla u \cdot \nabla v dx \\ \langle G'[u], v \rangle &= (p+1) \int_{\Omega} |u|^{p+1} uv dx. \end{aligned}$$

By the Lagrange multiplier rule we find some $\mu \in \mathbb{R}$ such that

$$\langle E'[\underline{u}] - \mu G'[\underline{u}], v \rangle = 0 \quad \text{for any } v \in H_0^1(\Omega).$$

Choosing $v = \underline{u}$ we find that

$$(p+1)\mu \int_{\Omega} |\underline{u}|^{p+1} dx = \int_{\Omega} |\nabla \underline{u}|^2 dx \geq 0.$$

Since $\underline{u} \in M$, $u \not\equiv 0$ so $\mu > 0$. Rescaling by $u = ((p+1)\underline{u})^{\frac{1}{p-1}} \in H_0^1(\Omega)$, we obtain

$$\int_{\Omega} \nabla u \cdot \nabla v - |u|^{p+1} uv dx = 0 \quad \text{for all } v \in H_0^1(\Omega)$$

i.e., $u \geq 0$ is a weak solution to

$$\begin{cases} -\Delta u = |u|^{p-1} u & \text{in } \Omega, \\ u \geq 0 & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

4. Bootstrap regularity. From $u \in H_0^1 \hookrightarrow L^{2^*}$,

$$-\Delta u = u^p \in L^{q_0}, \quad \text{where } q_0 = \frac{2^*}{p} > 1$$

By $W^{2,q}$ estimates

$$u \in W^{2,q_0} \hookrightarrow L^{q_1} \quad \text{where } q_1 = \frac{q_0}{n-2q_0} > q_0.$$

Repeating,

$$u \in W^{2,q} \hookrightarrow L^{q_2} \quad \text{where } q_2 = \frac{q_1 n}{n-2q_1} > \frac{n}{n-2q_0} q_1 = \left(\frac{n}{n-2q_0} \right)^2 q_0.$$

We iterate this procedure and obtain with $\left(\frac{n}{n-2q_0} \right)^k \rightarrow +\infty$ as $k \rightarrow +\infty$, we reach

$$u \in W^{2,q} \quad \text{with } q > n$$

in finitely many steps. Then Morrey inequality implies $u \in C^{1,\beta}$ for $\beta \in (0,1)$. In particular $u^p \in C^\alpha$, finally from Schauder we deduce

$$u \in C^{2,\alpha} \quad \text{for } \alpha \in (0,1).$$

5. Since $-\Delta u = |u|^{p-1} u = |u|^p \geq 0$ and $u \not\equiv 0$, by strong maximum principle for superharmonic functions, u cannot have an interior minimum, so

$$u > 0 \quad \text{in } \Omega.$$

This concludes the proof. ♣